World Health Organization

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Abbreviations

BGM	Blood glucose meter
BPMD	Blood pressure measurement device
EDL	Who model list of essential in vitro diagnostics
FIND	Foundation for Innovative New Diagnostics
HbA1c	Glycated haemoglobin
HIC	High income country
LMIC	Low and middle-income country
NCD	Noncommunicable disease
SMBG	Self-monitoring blood glucose
UHC	Universal health coverage
WHA	World Health Assembly
WHO	World Health Organization

Executive summary

Worldwide, more than 820 million people live with diabetes. Diabetes is among the top 10 causes of death and an increase in premature deaths in adults aged 30–69 years of 3% has been observed from 2000 to 2019. Medical devices, including in vitro diagnostics, such as blood glucose meters, the associated strips and lancets, syringes, and laboratory equipment to measure blood glucose and glycated haemoglobin are an integral part of the management of diabetes. For instance, syringes are essential for the administration of insulin, and blood glucose monitoring both in health facilities and at home are a cornerstone of proper diabetes management. A blood glucose meter is required for the successful management of diabetes which in turn prevents acute and chronic diabetes related complications. However, studies have shown limited availability of medical devices, including in vitro diagnostics, in low and middle-income countries. The Lancet Commission on Diagnostics highlighted that access to diagnostics is an essential component of achieving Universal Health Coverage by 2030. The Commission highlights a diagnostic gap of 56% for diabetes which is estimated to have a larger impact on morbidity and mortality than lack of access to treatment.

In recent years, actions to address the existing gaps in access to medicines and health technologies for diabetes management led to the launch of the Global Diabetes Compact by the World Health Organization. One of the aims within the Global Diabetes Compact is to improve access to diagnostics through workstream one, dedicated to essential diabetes medicines and associated health technologies. In addition, the global coverage targets for diabetes, include "100% of people with type 1 diabetes have access to affordable insulin and blood glucose self-monitoring". In 2021, the World Health Assembly adopted Resolution 74.4, Reducing the burden of noncommunicable diseases through strengthening prevention and control of diabetes which calls on Member States to ensure "... better access to safe, affordable, effective and quality essential diagnostics, medicines, and other related health products". Also, a project entitled "Improving access to Noncommunicable disease medicines and health technologies through improved assessment, regulation, prequalification, availability, price transparency, procurement and engagement with the private sector 2022-2024" was funded by the Norwegian Agency for Development Cooperation. This report is part of this project which includes two main components, a scoping review and an online survey to collect data on barriers, price and affordability of different medical devices. The overall aim of this report was to document the availability, price and affordability of glucometers, strips, lancets, syringes, blood pressure measurement devices, and facility testing for blood glucose, blood pressure and glycated haemoglobin.

The scoping review collected relevant available information on the price and availability of blood glucose meters and test strips. The price compilation aimed to gather relevant information previously collected by third parties. The online survey was sent to biomedical engineers and key opinion leaders to investigate issues of availability and price for blood glucose meters, test strips, lancets, syringes, blood pressure measurement devices, glycated haemoglobin analysers at the health facility and the price of tests and devices to individuals.

With regards to availability of these medical technologies data from the review and online survey highlight a lack of availability of the necessary medical devices for the proper management of diabetes. In looking at prices one methodological challenge is that prices of medical devices for diabetes management are reported inconsistently across the literature. From the data collected on prices the main finding is the variability of prices for blood glucose meters and strips. Although not from a representative sample some manufacturers of blood glucose meters and strips were found to have more products present in different markets than others. In comparing the price of a blood glucose meter versus the strips the cost of the strips outweighs the cost of the actual meter. Prices for lancets and syringes were also variable although less than for meters and strips. The cost of equipment for measuring glycated haemoglobin and the consumables was high. Blood pressure measurement devices were also found to be relatively high priced and again variability of prices, overall, within countries and by brands was found. When tests were performed in health facilities in some countries these are provided for free. If not, the price can be significant especially for glycated haemoglobin tests which is a cornerstone of diabetes management. In looking at the affordability of these different components of diabetes management the least affordable elements are strips and glycated haemoglobin testing meaning that many people will not be able to afford key aspects of blood glucose monitoring either daily at home or for regular follow-up in facilities.

In looking at the issue of availability, price, and affordability of medical devices there is the need to assess these factors from the perspective of the health system and the individual. The current market for blood glucose meters makes this unaffordable for people in low- and middle-income countries and new approaches to self-monitoring adapted to these contexts are needed. This might include the development of universal meters or strips. The data presented in this report highlights the wide range of prices for different medical devices and therefore price transparency is needed. The inclusion of blood glucose meters in World Health Organization prequalification, a service provided by the World Health Organization to assess the quality, safety and efficacy of medicinal products and devices, is an important step from a regulatory perspective, From the financing angle there is the need to clarify how medical devices can be included in Universal Health Care packages as the current prices are unaffordable for many people in low- and middle-income countries. Affordability of self-monitoring is linked to the recommended number of tests per day an individual need to successfully manage diabetes. Thus, the financial implications of the clinical guidance needs to be considered. To improve affordability, mechanisms need to be in place to improve the procurement and supply of medical devices especially ensuring linkages between the durable and consumable elements. Overall, there is a need to reconsider the element of affordability for any medical device for the management of diabetes where there is a durable and consumable element. In addition, any new technology which may eliminate the need for a consumable element will need to consider the ability to pay for people in low- and middle-income countries. Barriers identified in the literature comprise both international and national barriers. International challenges included high level barriers related to production and manufacturing, as well as accuracy and quality assurance issues. National barriers included issues within national health systems. Individual barriers include all those encountered by the individual while seeking care.

The Global Diabetes Compact, World Health Assembly Resolution 74.4, World Health Assembly Resolution 76.5 all mention the issue of availability and affordability of medical devices. With the target of “100% of people with type 1 diabetes have access to affordable insulin and blood glucose self-monitoring” being far from a reality in many contexts the World Health Organization, Member States, and other partners now need to find ways to ensure improved availability and affordability of these key elements for the management of diabetes to ensure improved outcomes for the millions of people with this condition.

1. Introduction

Trecia Simone Stewart,a certified emergency nurse takes the vital signs of her patient in Spanish Town, Jamaica, 2019 © WHO/Jayme Gershen

One of the five Noncommunicable diseases (NCD) prioritized by the World Health Organization (WHO) is diabetes. Worldwide, more than 820 million people live with diabetes *(1).* Diabetes is among the top 10 causes of death and is the only major NCD for which an increase in premature deaths in adults aged 30–69 years of 3% has been observed between 2000 and 2019 *(2)*. Globally it is estimated that 9 million people have type 1 diabetes *(3)* and about 60 million people with type 2 diabetes use insulin *(4).* The burden of diabetes-related morbidity and mortality in low- and middle-income countries (LMIC) is present and currently placing an extreme challenge on health systems, which are unable to provide insulin and the necessary medical devices for a variety of reasons. LMICs will also see the largest increase in the coming decades, thus requiring issues of access to medical devices to be addressed urgently.

Syringes are an essential medical product for the administration of insulin and blood glucose monitoring both in health facilities and at home are a cornerstone of proper diabetes management. A blood glucose meter (BGM) is used to monitor and track blood glucose levels. Whereas a rapid increase in technology has been seen in high income countries (HIC) for both the delivery of insulin and measurement of blood glucose, these innovations have not reached many in LMICs.

The World Health Organization (WHO) priority medical devices list (MDL) for management of cardiovascular diseases and diabetes *(5)* presents lists of all devices needed to diagnose and manage diabetes and its most common complications. The MDL list includes clinical interventions through levels of care and across the continuum of care. For example, this list states that HbA1c point of care testing should be available at Levels 2 and 3 of care, whereas blood pressure measurement devices (BPMD) and BGMs should be present at all levels of care. Particularly relevant to this report is the MDL inclusion of interventions for diabetes management including "Blood glucose management, including self-monitoring and insulin injections" which enumerates all medical devices needed for such intervention as follows: BGM, strips, syringes, needles, swabs, lancets and even the control solution to calibrate the BGM. Furthermore, all MDLs, including the one for cardiovascular diseases and diabetes, have been integrated in a database known as WHO Priority Medical Devices Information System (MeDevIS) *(6)* facilitating access to information of the devices. All medical devices in this publication are listed in MeDevIS. The WHO Model List of Essential In Vitro Diagnostics (EDL) provides a list of tests seen as essential for diabetes *(7).* These include blood glucose levels and haemoglobin A1c (HbA1c). For context with no laboratories, it recommends urine dipsticks and BGM for blood glucose levels and handheld analysers for HbA1c. When laboratories are available it recommends optical methods and automated chemistry analysers for blood glucose levels and immunoassays for HbA1c *(7).* From the wider perspective,

Studies have shown limited availability of different essential diagnostics in LMICs *(8).* The Lancet Commission on diagnostics *(9)* highlights that access to diagnostics is an essential component of achieving Universal Health Coverage (UHC) by 2030. For diabetes data provided by the Commission highlights a diagnostic gap of 56% which is estimated to have a larger impact on morbidity and mortality than lack of access to treatment. Lack of access to diagnostics is presented by the Commission as being caused by various factors including geographical access to diagnostics, socioeconomic factors of the population, lack of infrastructure, organizational factors within the health system and human resource capacity. Specifically for diabetes much attention has been paid to the issue of access to insulin *(10,11)*, as well as health system barriers to access to care and diagnostics *(12-15)*.

Responding to the increasing burden of diabetes around the world, the WHO has launched the Global Diabetes Compact. The WHO Global Diabetes Compact has the vision of reducing the risk of diabetes, and ensuring that all people who are diagnosed with diabetes have access to equitable, comprehensive, affordable and quality treatment and care *(16).* In order to achieve the objective, the Global Diabetes Compact has six workstreams. One of the workstreams is focused on improve access to essential diabetes medicine and associated heath technologies. This aspect of improving access to diagnostics is also reflected in the World Health Assembly (WHA) Resolution 74.4 with WHO calling on Member States to ensure "... better access to safe, affordable, effective and quality essential diagnostics, medicines, and other related health products" *(17)*. Another WHA Resolution 76.5 focused on diagnostics raised the overall issue of access to affordable and quality assured diagnostics with a specific mention on self-testing as well as the presence of diagnostics in facilities *(18)*. These high-level commitments have translated into a variety of concrete measures aimed to improve access to medical devices for diabetes with for example: "Asks" being put forward to medical device companies *(19)*; BGMs and HbA1c analysers being added to WHO Prequalification *(20)*; the inclusion of BGMs in the EDL for personal use *(21)*; as well as Global coverage targets for diabetes, including "100% of people with type 1 diabetes have access to affordable insulin and blood glucose self-monitoring" *(22)*.

To improve availability and access to medical devices, Norwegian Agency for Development Cooperation (NORAD) funded a project entitled "Improving access to NCD medicines and health technologies through improved assessment, regulation, prequalification, availability, price transparency, procurement and engagement with the private sector 2022-2024". This project is a part of the Global Diabetes Compact workstream one and this report is part of this overall project.

The aim of this report is to present new data from a scoping review of the literature, a price compilation and an online survey carried out by WHO to highlight the challenges with regards to access to BGMs, BGM strips, lancets, insulin syringes, HbA1c analysers and reagents (for facilities), BPMD, facility blood glucose testing and facility HbA1c testing for the management of diabetes to propose recommendations to improve knowledge of this issue and concrete steps to improve access.

2. Methods

Alusine, a patient with type 1 diabetes, checks his glucose levels with a glucometer provided by the noncommunicable diseases (NCD) clinic at Koidu Government Hospital, Sierra Leone, 2024 © WHO/Michael Duff

Figure 1 below presents the overall methodology, including the scope and data collection. The activities within the study were conducted sequentially, starting with the scoping review during the last quarter of 2022, followed by the design of the surveys in early 2023. Data was collected from June until October 2023, and the analysis and report writing were completed from January to June 2024. The table below outlines the various sources and data collected (Table 1).

Fig. 1. Methodology and collected data

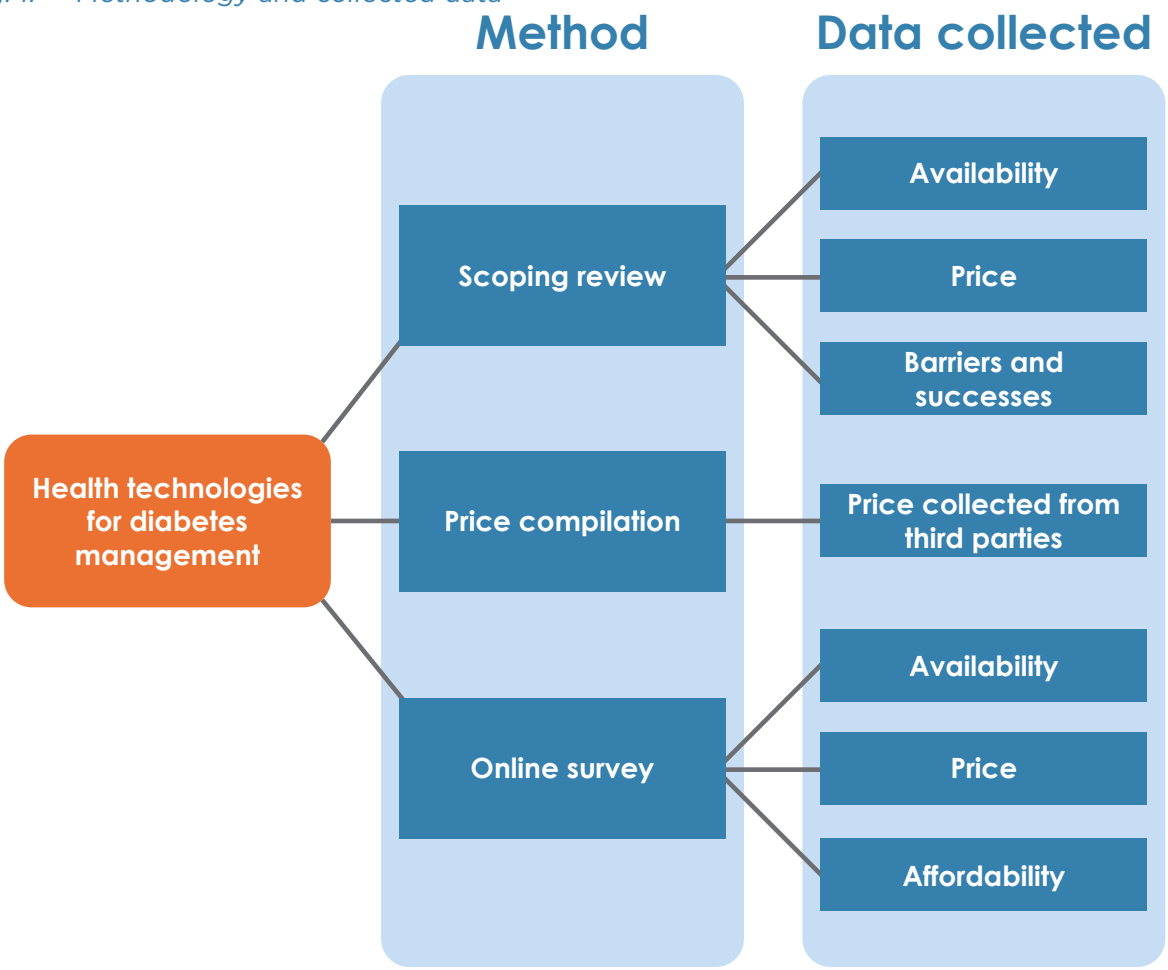


Table 1. Data on availability, price and affordability collected for different medical devices and laboratory tests and the source

Medical device/	Availability		Price		Affordability	
	Scoping review	Online survey	Scoping review	Online survey	Scoping review	Online survey
BGM	X	X	X	X	X	X
Strips for BGMs	X	X	X	X	X	X
Lancets		X		X		X
Insulin Syringes		X		X		X
HbA1c analysers and reagents (for facilities)		X		X		
BPMD		X		X		X
Facility blood glucose testing		X		X		X
Facility HbA1c testing		X		X		X

Prices were also compiled by searching for previously collected data (publicly available and shared by third parties).

2.1 Scoping review

The overall aim of the scoping review was to systematically collect relevant available information on medical devices for diabetes management. The scoping review was carried out using Google Scholar (search engine for scientific literature) for materials published in English only from 1990 onwards. Google Scholar was chosen for simplicity since it is open access. The initial search included the following search terms: "medical devices pricing", "medical devices diabetes pricing", "medical devices diabetes cost", "medical devices diabetes availability", "glucometer pricing", "glucometer cost" and "glucometer availability". We included publications in all types of diabetes mellitus (Type 1, Type 2, and gestational diabetes). This was done for all country income groups. All abstracts were reviewed by two people. The objective of the scoping review was to identify the following information in the literature:

1. Barriers and challenges for accessing the technologies for the management of diabetes.
2. Availability, prices and costs associated to technology for diabetes management.
3. Viable methodologies including dissemination strategies for the surveys, historically used indicators and questionnaires.
4. Strategies, reimbursement mechanisms and programs of successful provision in LMICs.

Notes were compiled using Mendeley Reference Manager on the necessary data to be included in the review. The results of the scoping review present the overall context and barriers in parallel to price and affordability data.

2.2 Price compilation

Third parties were contacted by WHO and a request was made for price and availability data on BMG and test strips, the Foundation for Innovative and New Diagnostics (FIND) provided data on the prices of BGMs, lancets and strips for different countries.

2.3 Online Survey

Two online questionnaires were designed by WHO and sent to two different audiences: biomedical engineers and key opinion leaders to be completed. The questionnaires were disseminated using the WHO newsletter, posted on biomedical engineering networks, and sent to a purposive sample of experts. The online questionnaire for biomedical engineers investigated issues of availability and price for BGMs, test strips, lancets, syringes, BPMDs, HbA1c analysers for health facilities and while the one for opinion leaders focused on the price of medical devices and tests to individuals. Both questionnaires are included in Annex 1 and Annex 2.

During the data analysis as data was missing on the actual manufacturer of BGMs this information was added based on the model reported in the questionnaire. For exchange rates if purchase year was provided the average annual exchange rate for the year reported was used. If no year was provided the average exchange rate for 2023 was used. Exchange rates were obtained from https://fxtop.com and prices were not adjusted for inflation. If a range of prices was given the average price was used. This was done except for price data reported by biomedical engineers for syringes and HbA1c analysers where the two values of the range provided are presented. For Costa Rica no currency was provided by the key opinion leader responding. Prices in Costa Rican Colons were not feasible with prices for a BGM less than US$ 1, so it was assumed that prices were reported in US$. In presenting data by income group the World Bank grouping was used *(23)*. If data is reported by regions this is done using the WHO Regions. Lowest wages were searched for on the internet and may not always be comparable.

In comparing the prices of BGM with testing strip, it was assumed that the price of the meter was amortized over 24 months and that individuals used 3 strips per day. This enabled the durable versus variable cost to be calculated. To calculate affordability from the key opinion leader data the median price was used with the following assumptions presented in Table 2.

Table 2. Assumptions for affordability calculations

Medical device/Laboratory test	Timeframe for affordability calculation
BGM	24 months
Strips	3 strips per day
Lancets	1 per day
Syringes	1 per day
BPMD	36 months
Tests (blood glucose, HbA1c)	per test

3. Results

World Diabetes Day, Brazzaville Congo 2019 © WHO/Marta Villa Monge

The results present the findings from the scoping review, data provided by FIND and the results of the online survey for both biomedical engineers and key opinion leaders.

3.1 Scoping review

Results from "search terms" were scanned for relevant titles. Abstracts of publications with relevant titles were scanned *(71)* and a subset of articles *(48)* was selected for full article review. Additional articles were identified via the references of the fully screened articles and were also included in the review *(23)*. A summary of all relevant data identified in each publication is listed in Annex 3.

Among the publications listing the price of the BGMs and/or the strips, some listed exact prices, while other publications listed price ranges *(24-27)* or prices aggregated by region or income level *(24,25,28)*. In contrast, some publications reported annual cost of BGMs and/or strips per person *(12,24,29,30)* rather than the price of the device itself. Other publications reported the cost of self-monitoring blood glucose (SMBG) including the combined cost of BGMs, lancets and strips. Annual cost of SMBG per person *(31-33)* or per population *(34)* were also reported and one publication reported the cost of SMBG per patient during pregnancy *(35)*. Another publication reported the annual median of out-of-pocket expenditure on medicines and medical devices combined *(26)*.

The most common period to report cost and expenditure was annually; however, expenses were also reported in monthly and daily intervals. One publication reported monthly expenses on medical devices and monthly expenses on strips. Another publication reported SMBG monthly cost *(36)*, and one publication reported the daily cost of strips *(28)*. An additional metric used across reviewed literature for expenditure on diabetes management included the percentage of income spent annually on diabetes care *(15)* (consultation costs, travel, insulin and SMBG) and percentage of income spent annually on BGMs and strips *(37)*. The cost of the fee associated with the service (blood glucose test) at the health facility *(27,38-44)* was reported mainly in settings where SMBG is not available. One publication reported the annual cost of blood glucose testing *(44)*.

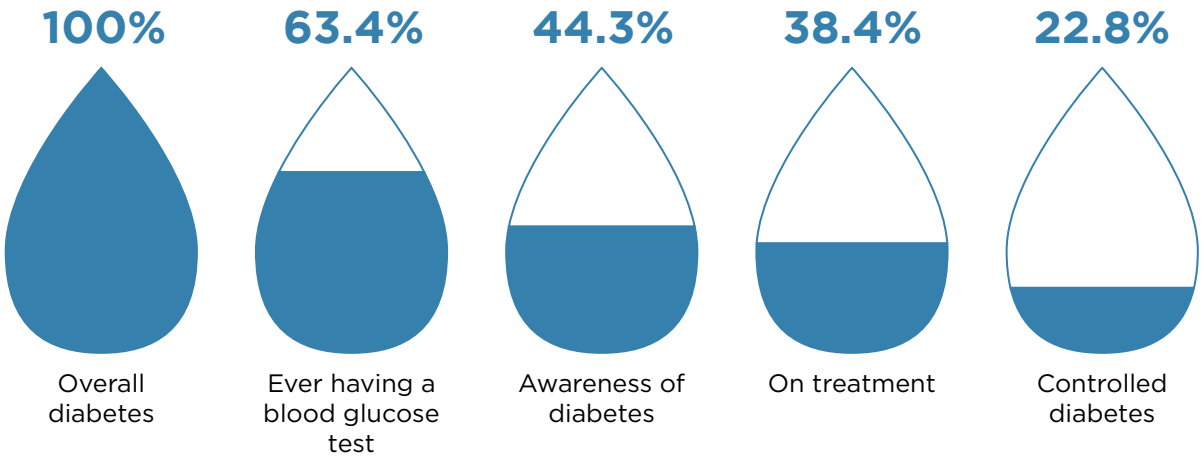
The literature also reported calculations of annual cost of managing diabetes per person *(30,32,33,38,45-47)* or for whole populations *(30-32,34)*. Calculations across the literature disaggregate the cost of diabetes in direct and indirect costs. The cost of medical devices was usually included within direct costs of diabetes. However, it is not always specified across disaggregated data and on occasion this was grouped with other costs. Annual direct cost of diabetes per person *(28,30,33,46,47)* and per population *(34,46)* identified within this review included components labelled as any of the following: daily self-management and home healthcare; medical devices and home healthcare; treatment and monitoring; BGMs and strips; and medical supplies. Annual excess cost of diabetes per person *(30,34)* and per population *(30,45,46)* were also reported across the literature. One publication reported excess cost per patient during pregnancy *(35)*, and one publication reported annual excess on medical devices specifically *(45)*.

Many studies report across diverse settings that the cost of SMBG exceeds the cost of insulin *(12,24,28-30,33,34,36,48,49)*.

3.1.1 Scoping review – Challenges and barriers to access to medical devices for the management of diabetes

Different barriers and challenges in the provision of diabetes care were commonly described in the literature. A total of 31 articles mention various barriers and challenges accessing diabetes care *(12,15,25,27,28,39-42,48,50-67)*. These barriers impact people being diagnosed with diabetes, receiving treatment and being controlled for diabetes. The cascade of care presented in Figure 2 shows that of all people with diabetes, only 63.4% have ever had a blood glucose test, 44.3% were aware of having diabetes, 38.4% of people were on some form of treatment, and 22.8% were defined as having controlled diabetes (HbA1c < 8.0% or equivalent) *(68)*. These data highlight the overall barrier of diagnosis and management of diabetes in LMICs due to lack of access to medical devices as well as weak health systems for the management of diabetes in LMICs.

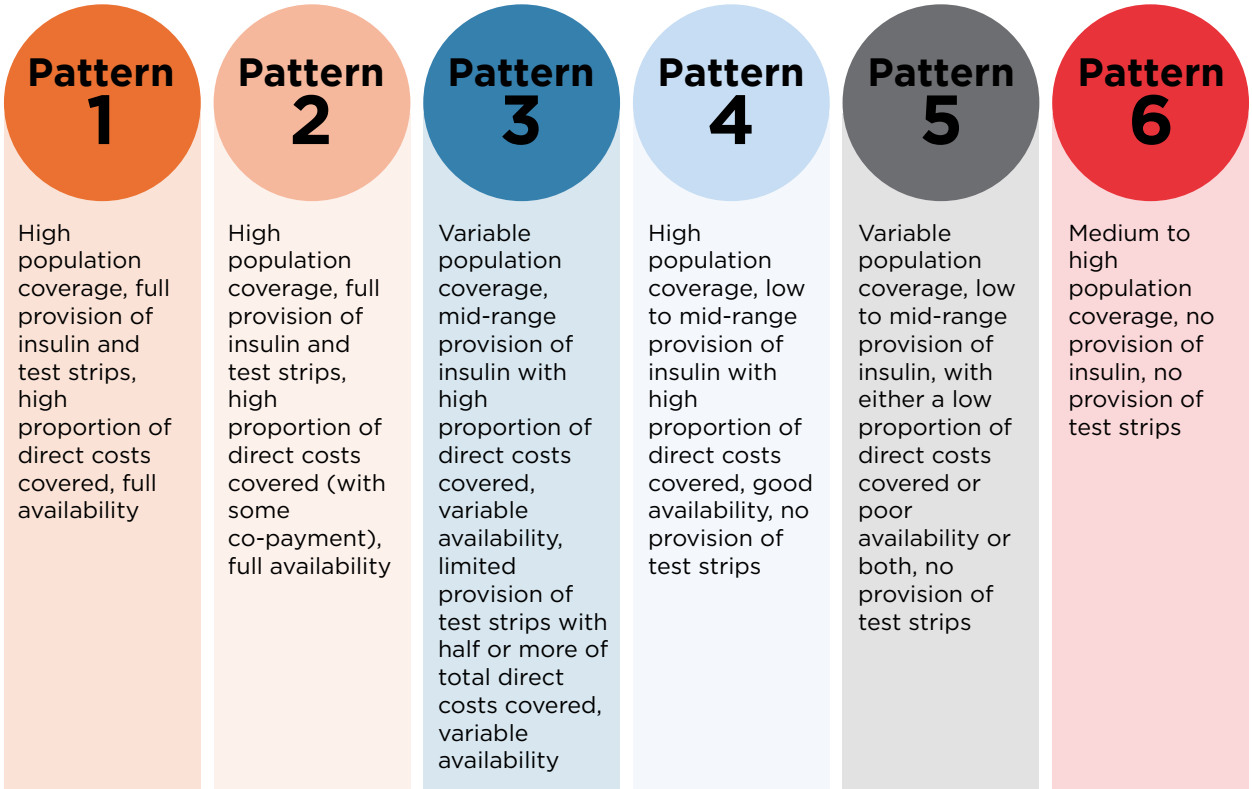
Fig. 2. Cascade of care for diabetes in LMICs (N=28)



Source: Adapted from Health system performance for people with diabetes in 28 low- and middle-income countries: A cross-sectional study of nationally representative surveys *(68)*.

Other challenges were identified in the literature review ranging from universal health coverage to revenue streams of the private sector. Klatman et al *(48)*, in a multi-country study including 44 countries, looked at three dimensions of universal healthcare (population coverage, service availability and direct cost) for the elements needed to manage type 1 diabetes. And identified six broad patterns of access with regards to UHC. These are presented in Figure 3.

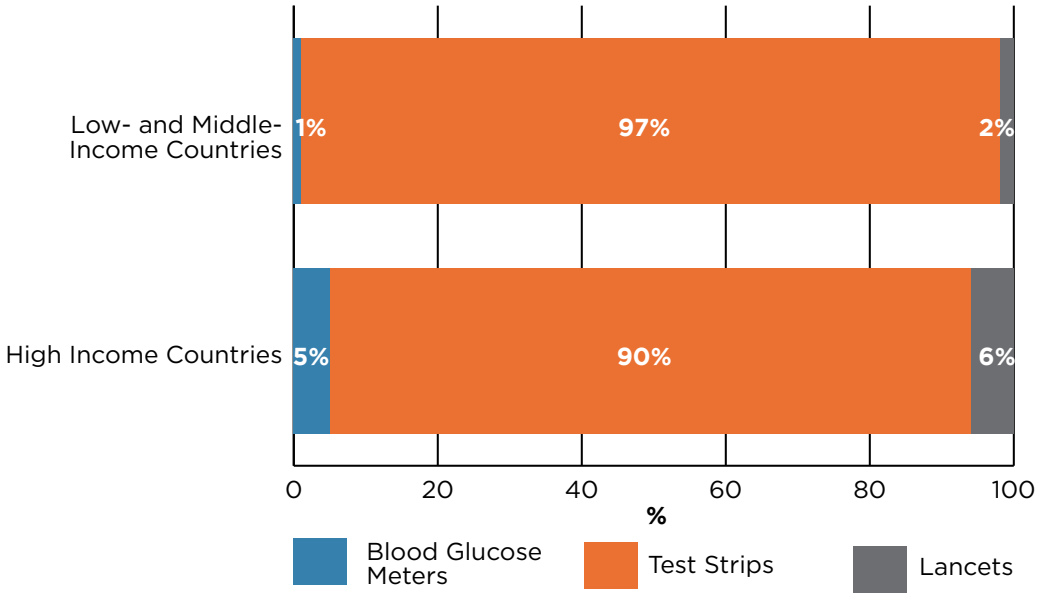
Fig. 3. Patterns of UHC and test strips



Source: Adapted from ocumenting and visualising progress towards Universal Health Coverage of insulin and blood glucose test strips for people with diabetes *(48)*.

The literature review also identified that the main revenue for the medical device industry was the test strips with these representing 90% of revenue in HICs and 97% of revenue in LMICs, in comparison to 5% and 1% respectively for the meters *(53)*. (Figure 4)

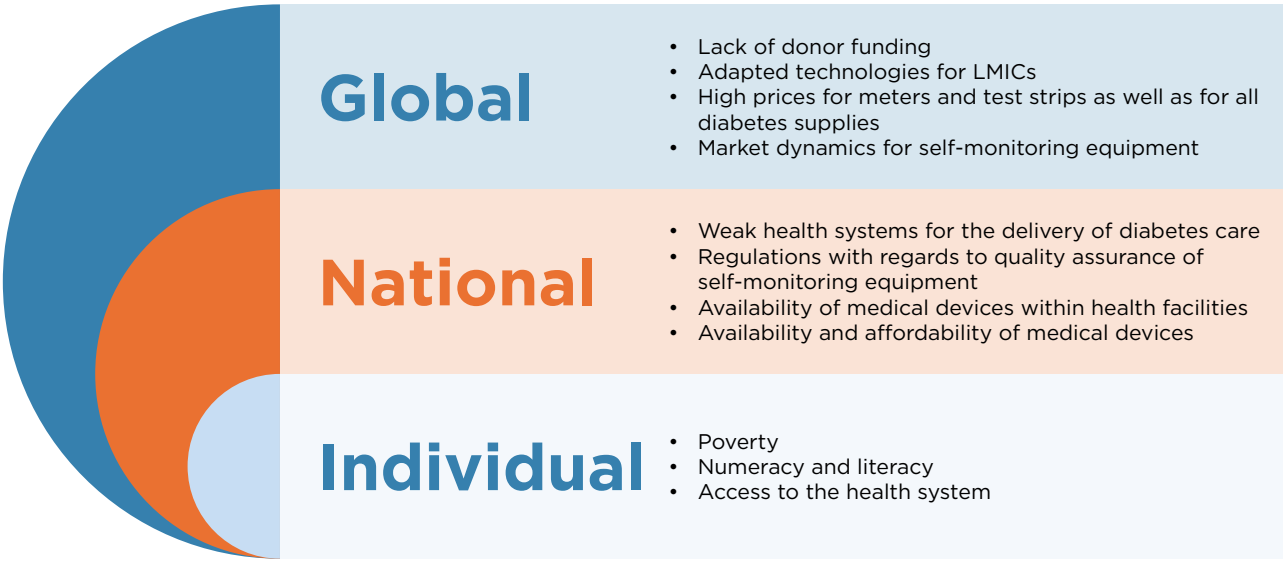
Fig. 4. Revenue split for different SMBG commodities



Source: Adapted from Market report. Diabetes self-monitoring devices in low-and middle-income countries *(53)*.

A summary of barriers is presented, below dividing these into international barriers including high level barriers related to production and manufacturing, as well as accuracy and quality assurance issues. National barriers include issues within national health systems. Individual barriers include all those encountered by the individual while seeking care.

Fig. 5. Challenges and barriers to access to medical devices for the management of diabetes



3.1.2 Scoping review – Availability of medical devices for the management of diabetes

Throughout the literature availability was reported in different scales (regional, national, facility, and individual) and with different metrics. Availability was mostly presented as percentages, but on occasions it was reported using a qualitative scale (e.g., mostly available, rarely available, never available). Previous WHO publications reported availability of BGMs (along with other essential health technologies) by WHO Region and by income level *(69)*. Different levels of availability use different metrics. For instance, at regional level, it is the percentage of countries with the technology available; at a national level, it is a percentage of facilities with the technology available; at the patient level is the proportion of consultations when the technology was available. In WHO service availability and readiness assessment (SARA), service availability refers to the physical presence of the delivery of services, encompassing health infrastructure, core health personnel, and service utilization.

3.1.3 Scoping review – Price of BGMs

Prices of medical devices for diabetes management were reported inconsistently across the literature and brands and models were never mentioned. Therefore, comparing prices across publications is often not possible. Prices were mostly reported in US Dollars, but some publications included cost in local currencies. Beside the price of the device, in settings where SMBG is not commonly available the patient fee for facility-based testing was reported. Other commonly used metrics included the cost of diabetes for patients and country populations. Most cost calculations included patients' direct cost (e.g., medicines, consultation, hospitalization) and indirect cost (e.g., transportation cost, days of work lost). In some calculations the expenses of people with diabetes were often compared against people without diabetes to estimate the excess cost of diabetes. In some studies models using secondary data were created to estimate countries' annual expenditure in diabetes. Public data was also used to estimate national diabetes expenditure.

From the review of the literature eight publications reported the unit price for a BGM for seven countries *(28,35,39-41,53,54,70)*. The prices reported ranged from US$ 9.00 to US$ 131.3.

Table 3. Prices of BGMs from the scoping review

Country	BGM reported price (US$)	Year of publication
Kenya *(54)*	$5.90	2016
Mexico *(35)*	$42.00	2019
Mozambique *(41)*	$104.86	2009
Netherlands (Kingdom of the) *(28)*	$44.10	2015
Nicaragua *(39)*	$120.00	2007
United Republic of Tanzania *(28)*	$12.90	2015
Viet Nam *(40)*	$87.03	2009
Low-income countries *(53)*	$9.00–25.00	2021
High-income countries *(53)*	$10.00–45.00	2021

3.1.4 Scoping review – Price of strips for BGMs

Fourteen publications found in the literature review reported the price of strips for BGMs for 70 countries with some countries providing strips for free to over US$ 1 per strip *(24-28,31,39-41,45,48,52-54,70)*. All the data on the price per strip is presented in Table 4.

Table 4. Prices of strips from the scoping review

Country	Strip reported price (US$) per one strip	Year of publication
Albania; Israel; North Macedonia; Serbia	$0.00–0.67	2013
Australia *(28)*	$0.40	2019
Austria; Belgium; France; Germany; Ireland; Luxembourg; Netherlands (Kingdom of the); Sweden; The United Kingdom *(26)*	$0.00–0.84	2013
Azerbaijan; Georgia; Russian Federation; Ukraine *(26)*	$0.00–0.65	2013
Azerbaijan *(28)*	$0.70	2019
Bangladesh *(28)*	$0.27	2019
Bangladesh *(70)*	$0.50	1999
Bolivia (Plurinational State of) *(28)*	$0.62	2019
Bulgaria; Czechia; Latvia; Lithuania; Poland; Romania *(26)*	$0.00–0.63	2013
Burkina Faso *(28)*	$0.54	2019
Burundi *(28)*	$0.38	2019
Cambodia *(28)*	$0.30	2019
Central African Republic *(28)*	$1.00	2019
Central and Eastern Europe *(25)*	$0.28–1.14	2000
Democratic Republic of the Congo *(28)*	$0.50	2019
Congo *(28)*	$0.32	2019
Croatia; Greece; Italy; Portugal; Slovenia; Spain *(26)*	$0.00–0.78	2013
Dominican Republic *(28)*	$0.82	2019
Ecuador *(28)*	$1.07	2019
Eritrea *(28)*	$0.59	2019
Ethiopia *(28)*	$0.65	2019
Ghana (National Health Insurance Scheme) *(28)*	$1.11	2019
Global average *(28)*	$0·29	2019
Global average *(24)*	$0.60–1.0	2015
Guatemala *(28)*	$0.48	2019
Guyana *(28)*	$0.48	2019
Haiti *(28)*	$0.51	2019
High-income countries *(53)*	$0.17–0.87	2021
Jamaica *(28)*	$0.54	2019
Japan *(28)*	$0.71	2019
Kenya (National Health Insurance Fund) *(28)*	$0.58	2019
Kenya *(54)*	$0.75	2016
Liberia *(28)*	$1.06	2019
Low-income countries *(53)*	$0.06–0.88	2021

Country	Strip reported price (US$) per one strip	Year of publication
Maldives *(28)*	$0.71	2019
Mali *(28)*	$1.65	2019
Mauritania *(28)*	$0.33	2019
Mexico (Social Security Institute) *(28)*	$0.38	2019
Mexico (Institute for Social Security and Services for State Workers) *(28)*	$0.38	2019
Mexico (Seguro Popular) *(28)*	$0.38	2019
Mozambique *(41)*	$0.59	2009
Nepal *(28)*	$0.56	2019
Nicaragua *(39)*	$0.20	2007
Nigeria *(28)*	$0.56	2019
Pakistan *(28)*	$0.25	2019
Philippines *(28)*	$0.39	2019
Rwanda (Mutuelles de Santé) *(28)*	$0.37	2019
Sri Lanka *(28)*	$0.33	2019
Sudan (Khartoum only) *(28)*	$0.36	2019
Tajikistan *(28)*	$0.45	2019
United Republic of Tanzania (National Health Insurance Fund) *(28)*	$0.88	2019
United Republic of Tanzania *(28)*	$0.88	2019
Togo *(28)*	$0.54	2019
Uganda *(28)*	$0.28	2019
The United Kingdom *(31)*	$0.39	2018
Uzbekistan *(28)*	$0.42	2019
Viet Nam *(28)*	$0.47	2019
Viet Nam *(40)*	$0.49	2009
Zambia *(27)*	$0.58-0.74	2004

3.2 Price compilation

Data aggregated from FIND on the prices of lancets is presented in Table 5.

Table 5. Prices of lancets from FIND

Country	Lancet reported price (US$) per lancet	Year
Brazil	$0.01–0.03	2020
South Africa	$0.17	2019
Ethiopia	$0.07	2019
United Republic of Tanzania	$0.08	2019
Bangladesh	$0.03	2019
India	$0.02	2019
Peru	$0.07	2019
Senegal	$0.06	2019

Data provided by FIND on the prices of BGMs in the private sector show a wide range of prices presented in Table 6.

Table 6. Prices of BGMs from FIND

County	BGM reported price (US$)	Year
Bangladesh	$12.99	2019
Ethiopia	$19.54	2019
India	$11.83	2019
Peru	$29.32	2019
Senegal	$23.42	2019
United Republic of Tanzania	$25.81	2019

FIND also provided some information on the price of test strips. (Table 7). As with price data on strips from the scoping review a wide range is seen across countries.

Table 7. Prices of strips from FIND

Country	Strip reported price (US$) per strip	Year
Bangladesh	$0.18	2019
Brazil	$0.32	2018
Brazil	$0.09–0.12	2018
Brazil	$0.08–0.10	2020
Ethiopia	$0.37-0.65	2018
Ethiopia	$0.07	2018
Ethiopia	$0.37	2019
India	$0.17	2019
Kenya	$0.26- 0.56	2020
Kenya	$0.24–0.73	2020
Kenya	$0.16–0.35	2020
Peru	$0.33	2019
Senegal	$1.26	2019
South Africa	$0.23–0.35	2018
South Africa	$0.06	2018
South Africa	$0.35	2019
United Republic of Tanzania	$0.41–0.88	2018
United Republic of Tanzania	$0.12	2018
United Republic of Tanzania	$0.41	2019
Ukraine	$0.22–0.34	2020
Ukraine	$0.29	2020

3.3 Online questionnaire

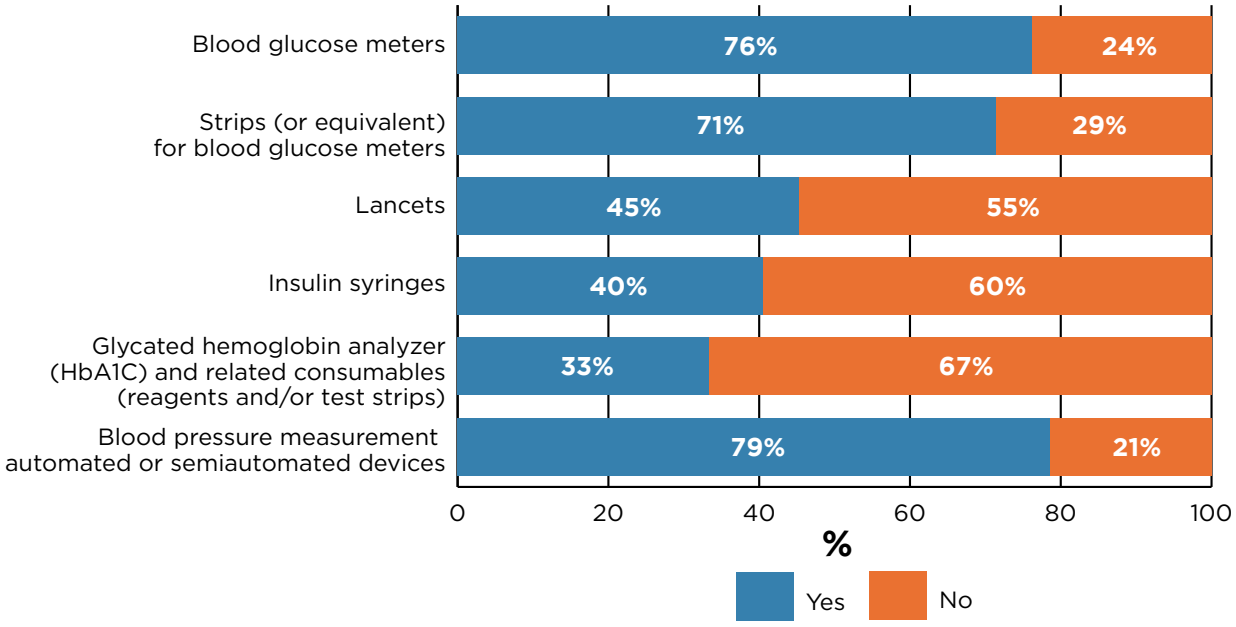
The results of the online questionnaire are presented separately for biomedical engineers and key opinion leaders.

3.3.1 Online questionnaire – biomedical engineers

For the online questionnaire a total of 120 records were submitted online by biomedical engineers. Of these 73 records contained no information. This left 47 records with some data to be analysed. From these responses, one respondent stated they could not provide any information). Therefore, the overall analysis of the results for biomedical engineers was based on 46 records with many questions not or only partially answered. Thirty-nine respondents provided their role/position. Of these, 31 were biomedical engineers in different roles; 6 mentioned senior management roles (e.g., Medical Director); 1 nurse and 1 student. The description of the respondents can be found in Annex 4.

With regards to providing procurement and pricing information different number of biomedical engineers stated they were able to provide this information (Figure 6). It should be noted that not all those who stated they could provide this information did.

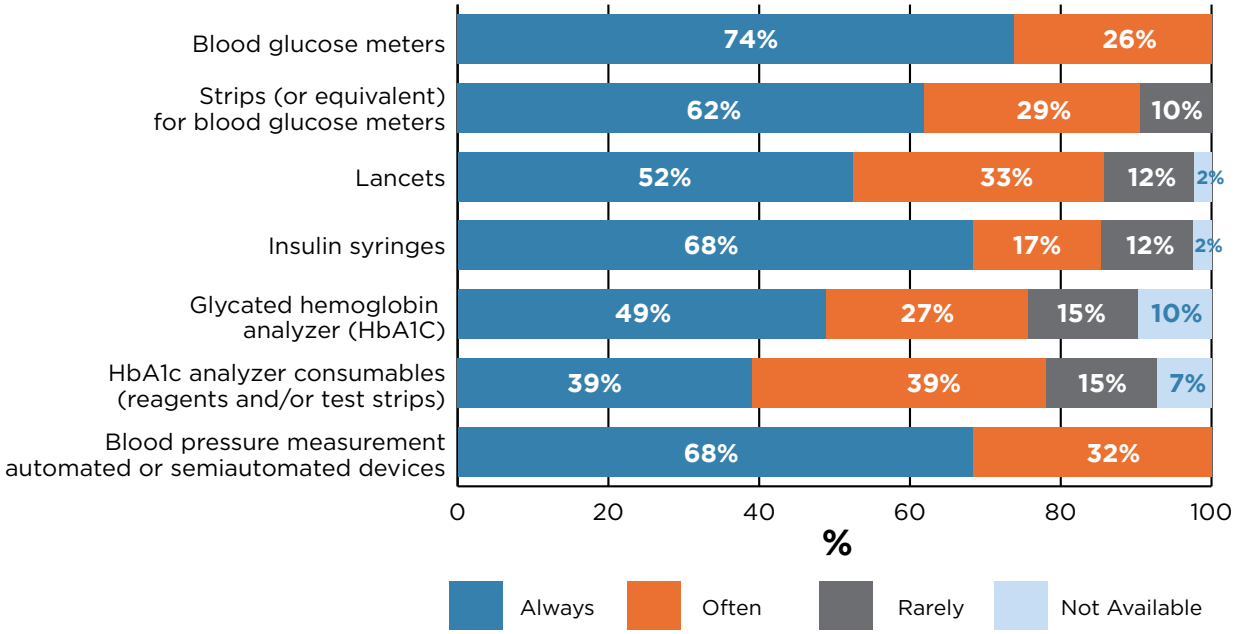
Fig. 6. Biomedical engineer respondents stating (percentage of total) that they could provide information on procurement and pricing of the different devices (N=46)



3.3.1.1 Biomedical engineers – availability of different medical devices

Biomedical engineers answers about the availability of the different medical devices at the health facility are presented in Figure 7.

Fig. 7. Availability of different devices at the health facility (percentage of respondents providing specific answer) (N=46)



From above it is interesting to see the difference in reported availability between the different devices and the difference between medical technologies and the reagents needed. No analysis was done looking at different facilities, countries or environments given the limited data available.

3.3.1.2 Biomedical engineers – Price of BGMs

The data provided by biomedical engineers on pack size and quantity purchased was often missing or hard to interpret. Therefore, only manufacturer, brand and price data were analysed. Respondents from five countries (Botswana, Kenya, Lebanon, Malawi and Nigeria) provided some information on manufacturers, brands, and prices of BGMs. Lebanon only provided the manufacturer and brand as well as one respondent for one type of BGM in Nigeria. Eleven different manufacturers of BGMs were reported as follows by the respondents. (Table 8)

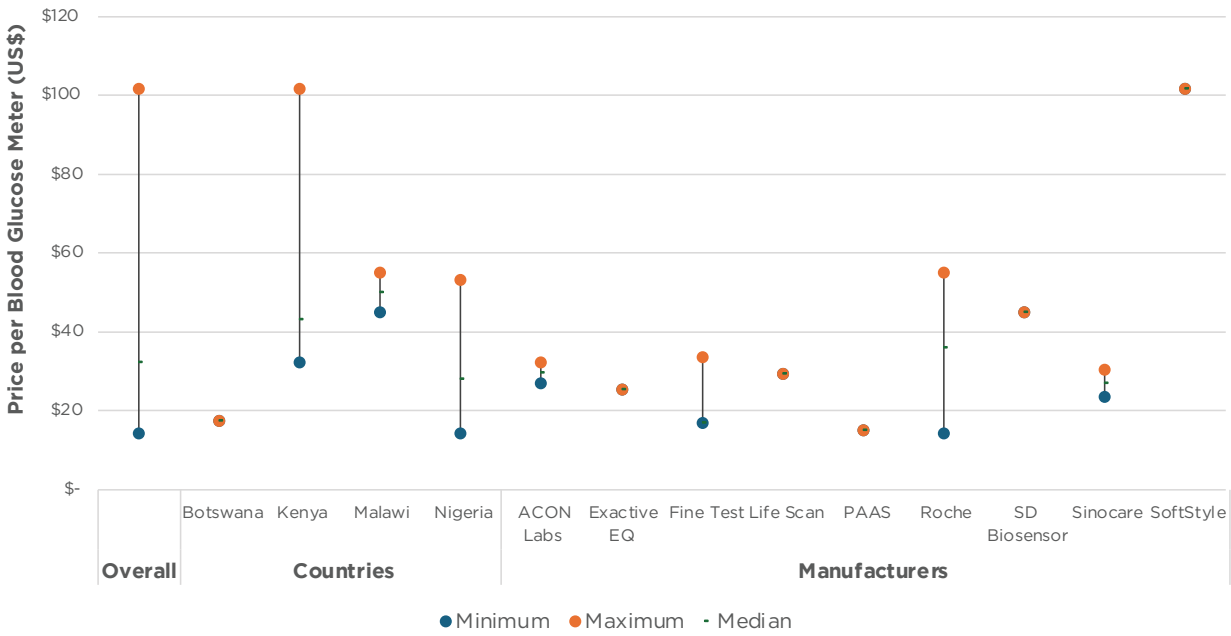
Table 8. Number of mentions of different brand of BGMs by country

Brand	Number of mentions	Countries
Roche	10	Botswana; Kenya (2 mentions); Malawi; Nigeria (6 mentions)
Fine test	3	Nigeria (3 mentions)
ACON Labs	2	Kenya; Nigeria
Sinocare	2	Nigeria (2 mentions)
Exactive EQ	1	Nigeria
Keto Mojo	1	Lebanon
Life Scan	1	Nigeria
PAAS	1	Nigeria
SD Biosensor	1	Malawi
Softstlye	1	Kenya

In looking at the manufacturers brands that were included in the responses for Roche all were Accu-Chek (with different models reported); for Fine test only one of the three responses mention that this was the Premium Brand; for ACON Labs these were the On Call Brand (with two different brands reported); and for Sinocare of the two responses one mentioned the Safe AQ Brand.

For price data 21 data points were available for analysis from 4 countries and 9 manufacturers. A wide range can be seen in the prices for BGMs overall, within countries and even within brands (see Fig.8.). The highest priced BGM overall was 7.1 times more expensive than the lowest priced.

Fig. 8. Minimum, maximum, and median prices for blood glucose meters as reported by bio-medical engineers



From the comments left by some biomedical engineers two themes emerge with regards to the prices of BGMs: 1. the prices are variable and not stable; and 2. the price of the glucose meter can depend on the purchase of strips and that in one case the meters can be given for free if the health facility agrees to purchase a certain quantity of strips per month.

3.3.1.3 Biomedical engineers – Price of BGM strips

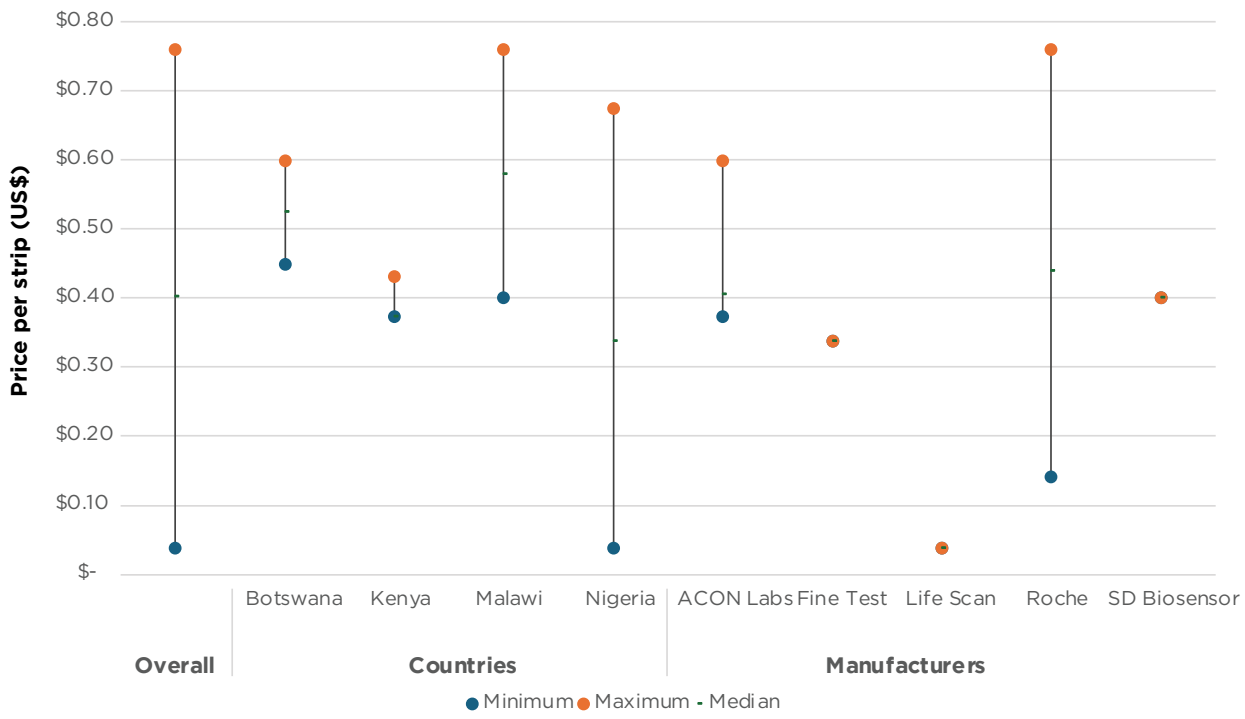
As with BGMs the data provided by biomedical engineers on pack size and quantity purchased was often missing or hard to interpret. Therefore, only manufacturer, brand and price data were analysed. Respondents from Botswana, Kenya, Malawi, and Nigeria provided some information on Manufacturers, Brands, and prices of strips for BGMs. Five different manufacturers of strips for BGMs were reported as follows.

Table 9. Number of mentions of different brand of BGM strips by country

Brand	Number of mentions	Countries
Roche	9	Botswana; Kenya (2 mentions); Malawi; Nigeria (5 mentions)
ACON Labs	3	Botswana, Kenya, Nigeria
Fine test	2	Nigeria (2 mentions)
Life Scan	1	Nigeria
SD Biosensor	1	Malawi

For price data 16 data points were available for analysis from 4 countries and 5 manufacturers. As with BGMs a wide range of prices is found overall, by country and by brand. (Figure 9) The highest priced test strip is 20.1 times more costly than the lowest priced strip.

Fig. 9. Minimum, maximum, and median prices for strips for BGMs as reported by biomedical engineers

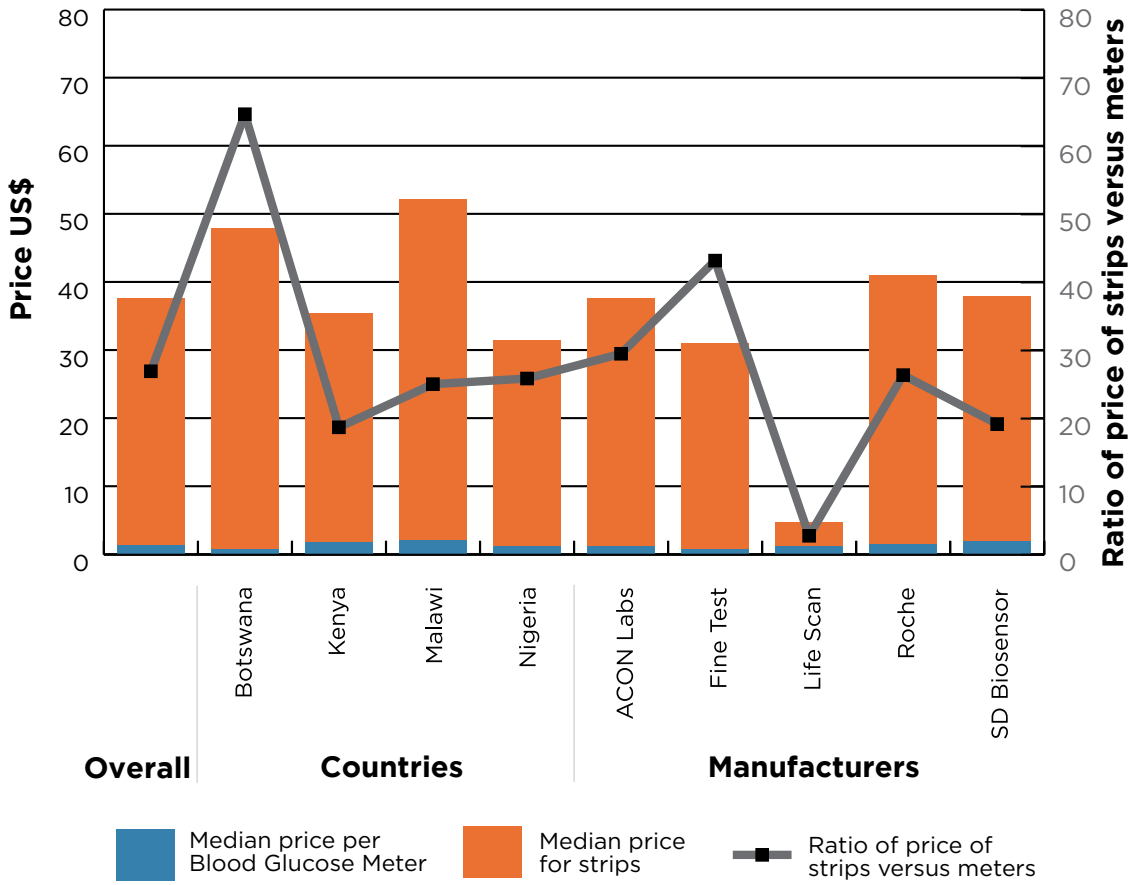


The variability of prices was highlighted within the comments left by respondents.

3.3.1.4 Biomedical engineers – Price of BGMs versus strips

As part of the analysis a comparison was done for the cost of the meter versus the cost of the strips as provided by biomedical engineers. In most cases, except for Life Scan the monthly cost of strips far outweighs the cost of the actual meter. (Figure 10)

Fig. 10. Monthly cost of Blood Glucose Meter versus Strips (assuming 3 strips per day) from data reported by biomedical engineers and ratio of price of strips to glucose meters



3.3.1.5 Biomedical engineers – Price of lancets

There was a dearth of data reported on lancets. The price of one type of lancet in Nigeria was US$ 0.03. One respondent stated there is instability of prices.

3.3.1.6 Biomedical engineers – Price of HbA1c analysers

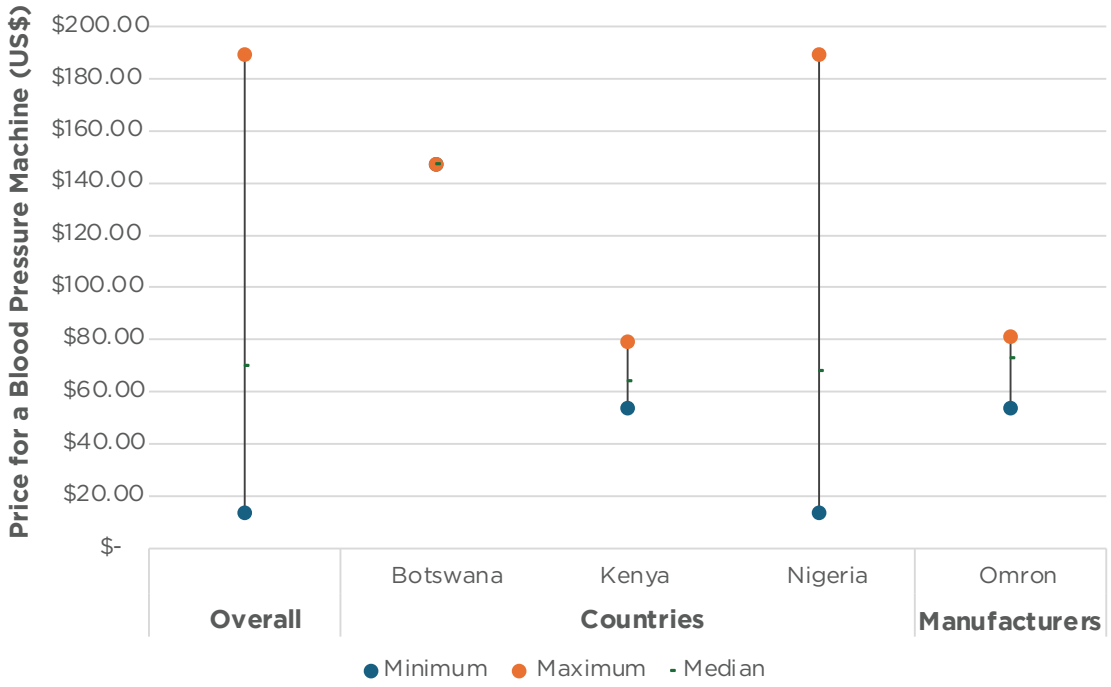
Only one biomedical engineer provided data for HbA1c analysers and their consumables and provided a range of prices for the machine (US$ 64.07-US$ 75.87) and for the consumables (US$ 67.44 -US$ 79.24).

3.3.1.7 Biomedical engineers – Price of BPMDs

The data provided by biomedical engineers on pack size and quantity purchased was often missing or hard to interpret. Therefore, only manufacturer, brand and price data were analysed. Respondents from five countries (Botswana, Ecuador, Kenya, and Nigeria) provided some information on manufacturers, brands, and prices of BPMDs. Ecuador provided all data except for the currency. This resulted in 13 data points for BPMDs. Of these 6 were for different Omron machines and the 8 remaining for various other brands.

The results from biomedical engineers on using these 13 prices from 3 countries and 7 manufacturers are presented in Figure 11. A wide range can be seen in the prices for BPMDs overall and within countries. The highest priced BPMDs was 14 times more expensive than the lowest priced.

Fig. 11. Minimum, maximum, and median prices for BPMDs reported by biomedical engineers



3.3.1.8 Biomedical engineers – Price of different laboratory tests

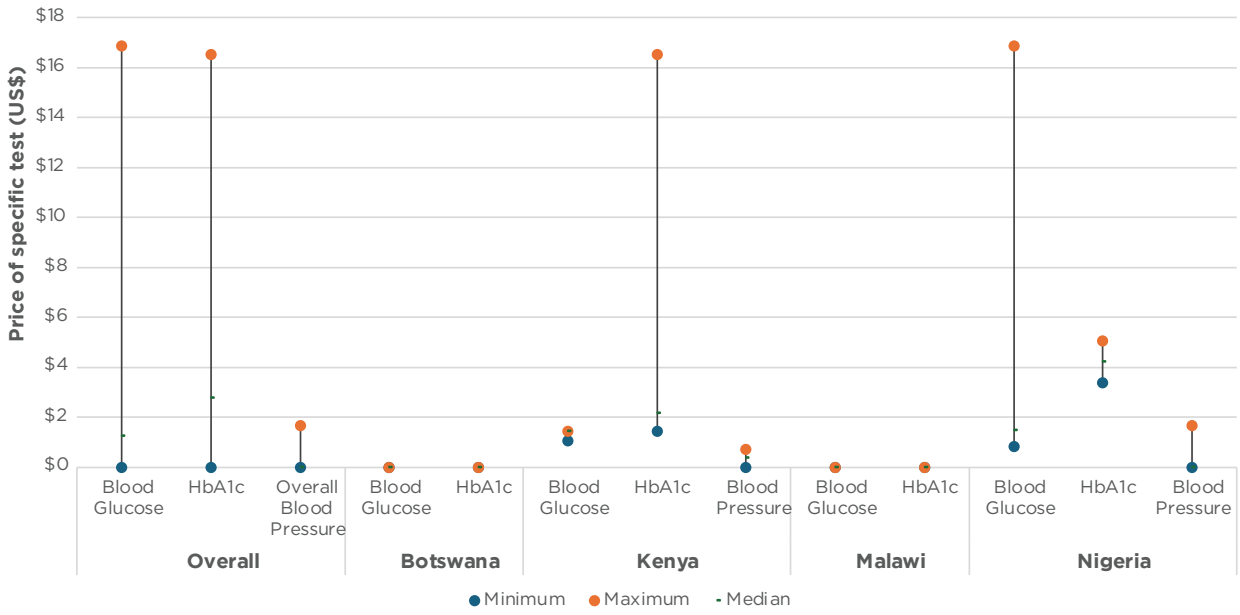
Table 10 shows the different prices that were provided by Biomedical engineers with regards to the prices of different tests for patients.

Table 10. Countries and number of data points for different tests done reported by biomedical engineers

Country	Blood Glucose	HbA1c	Blood Pressure
Botswana	1	1	0
Kenya	3	3	2
Malawi	1	1	0
Nigeria	7	4	5

In Botswana and Malawi blood glucose testing and HbA1c were reported as being free. Some respondents mentioned blood pressure measurements being free in Kenya and Nigeria. Overall, a wide range of prices for the different tests can be seen with this being pronounced in Nigeria with a wide range of prices for each test. (Figure 12) Due to lacking data no analysis could be done based on type of facility or if this was in the public or private sector.

Fig. 12. Minimum, maximum, and median prices for different tests as reported by biomedical engineers



3.3.1.9 Biomedical engineers – Affordability of medical devices for the management of diabetes

In looking at affordability with tests reported by biomedical engineers (blood glucose, HbA1c and blood pressure) being free in Botswana and Malawi these are of course affordable. The data from Kenya and Nigeria shows that many of the tests are not affordable for those on low wages in these countries.

Table 11. Price and affordability of different tests in Kenya and Nigeria as reported by biomedical engineers

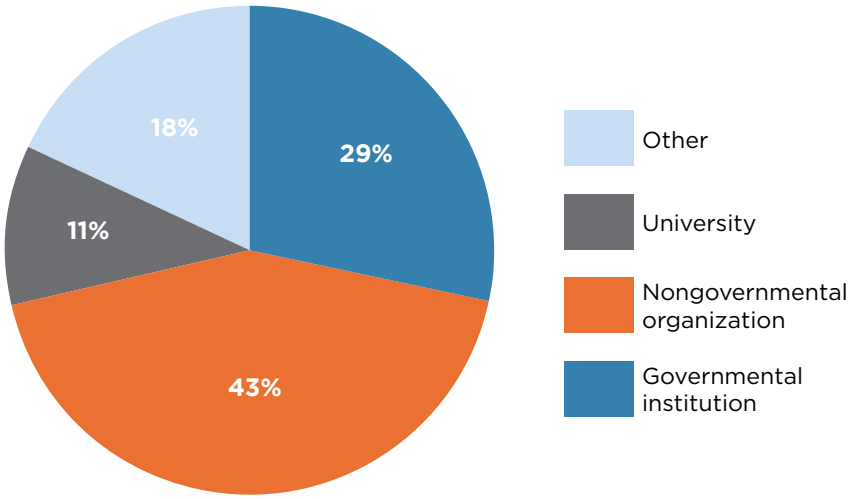
Country	Day's wage (US$)	Blood glucose test		HbA1c		Blood pressure	
		Price (US$)	Affordability (days wages)	Price (US$)	Affordability (days wages)	Price (US$)	Affordability (days wages)
Kenya	$3.18	$1.44	0.5	$1.44	0.5	$0	0.0
Kenya	$3.18	$1.44	0.5	$2.15	0.7		
Kenya	$3.18	$1.08	0.3	$16.51	5.2	$0.72	0.2
Nigeria	$1.24	$1.69	1.4	$4.22	3.4	$0	0.0
Nigeria	$1.24	$0.84	0.7	$3.37	2.7	$0	0.0
Nigeria	$1.24	$0.84	0.7			$0	0.0
Nigeria	$1.24	$2.53	2.0				
Nigeria	$1.24	$1.26	1.0				

3.3.2 Online questionnaire – key opinion leaders

A total of 154 records were submitted online by key opinion leaders. Of these 95 records contained no information. This left 65 records with some data to be analysed. Details of the respondents can be found in Annex 5.

Fifty-six respondents provided information as to the type of organization they worked at. Most respondents reported working for Nongovernmental organizations (Figure 13).

Fig. 13. Distribution of types of organizations respondents reported working at (N= 65)

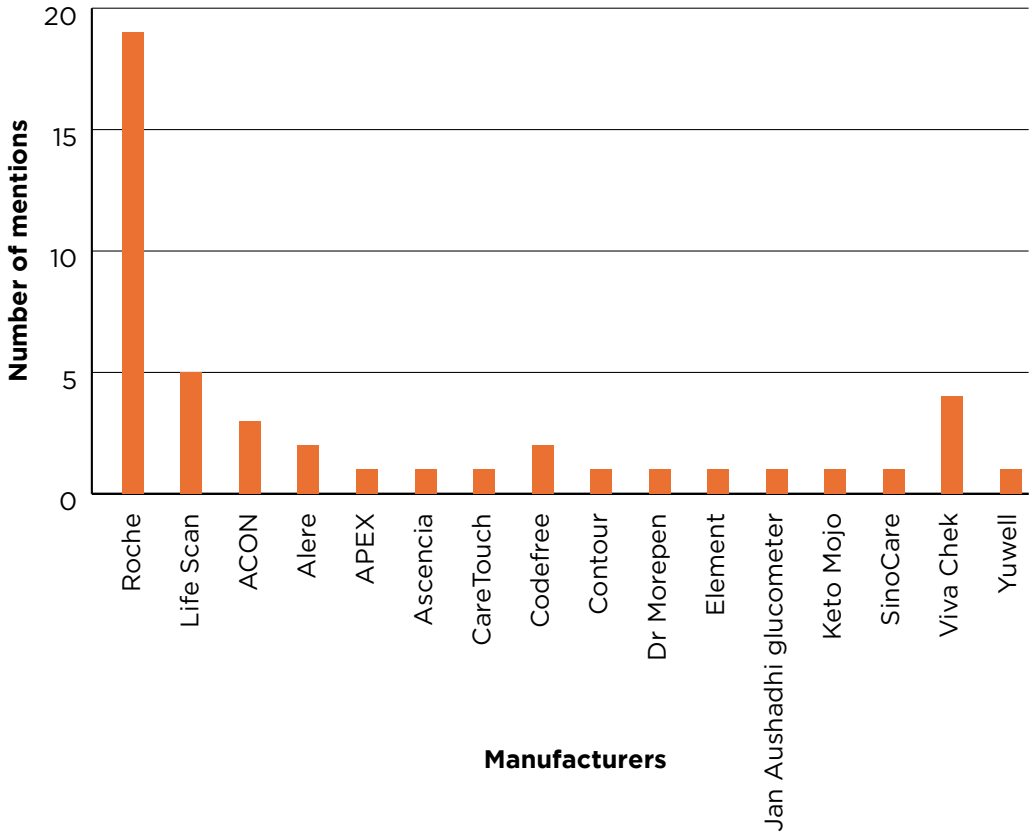


In the "Other" category this included people working at: hospitals; start-ups; a student; private sector; health insurance organization; and a diabetes association. Those reporting working for a governmental institution were either at the Ministry of Health or National Regulatory Agency. Thirty-seven respondents despite saying they would provide information on prices did not do this. Thus, the overall analysis of the results for key opinion leaders will be on 19 records with many questions not or partially answered.

3.3.2.1 Key opinion leaders – availability of BGMS for individuals

The data provided by key opinion leaders on pack size, facility type was often missing or hard to interpret. Therefore, only manufacturer, brand and price data were analysed. Respondents from 12 countries (Cameroon, Canada, Costa Rica, Cyprus, Eswatini, Guinea, India, Iraq, Latvia, Malta, Nigeria, South Africa, and Sweden) provided some information on Manufacturers, Brands, and prices of BGMs. Forty-five different manufacturers of BGMs were reported as presented in Fig.14.

Fig. 14. Number of mentions for each manufacturer of BGMs by key opinion leaders (N=19)



In terms of brands, for those manufacturers with more than one mention, all the models from: Roche were Accu-Chek (different models reported); Life Scan One Touch (different models reported); Vivia Chek Ino and Smart were reported; and ACON On Call (different models reported). For Alere and Codefree no clear models were reported.

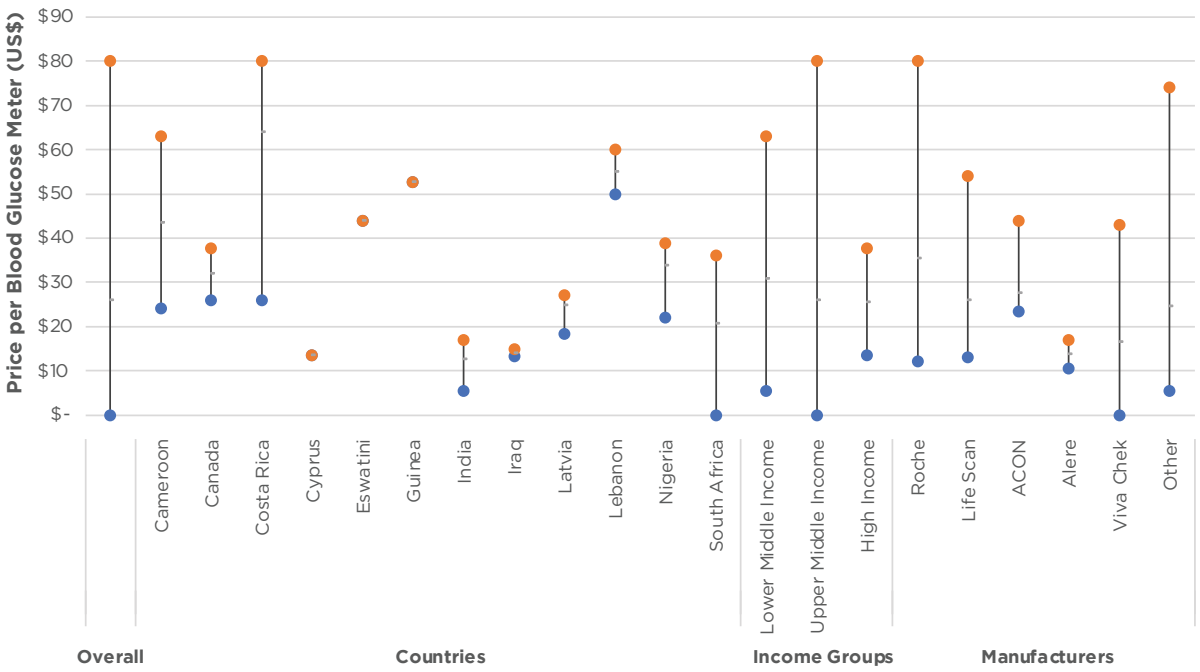
3.3.2.2 Key opinion leaders – Price of BGMs for individuals

For price data 45 data points were available for analysis from 12 countries and 16 manufacturers. A wide range can be seen in the prices for BGMs overall, within countries, within country income groups and even within brands. The highest price reported was in Costa Rica (US$ 80). It is important to note that the respondent from Costa Rica did not specified currency therefore USD were assumed. Looking at country income groups the median price was highest in lower middle-income countries (US$ 30.68), followed by upper middle-income countries (US$ 26.00) and finally high-income countries (US$ 25.40).

The respondent from Malta did not report any price data for BGMs but stated in the "Comment" section: "In the NHS (public service), BGMs are provided for free to all [people with diabetes]. Procurement for BGMs and compatible test strips is through a tendering system, and the brand and model may change according to chosen offers". A similar scheme should be available in South Africa since a price of US$ 0.00 was reported.

It is interesting to note that in both India and Sweden one source where people bought their BGMs, and Strips was in online stores.

Fig. 15. Minimum, maximum, and median prices for BGMs as reported by key opinion leaders



Comments from the respondents highlight the cost of BGMs for populations as well as the existence in some contexts of donation programs or that sometimes BGMs can be provided for free.

3.3.2.3 Key opinion leaders – Price of BGM strips

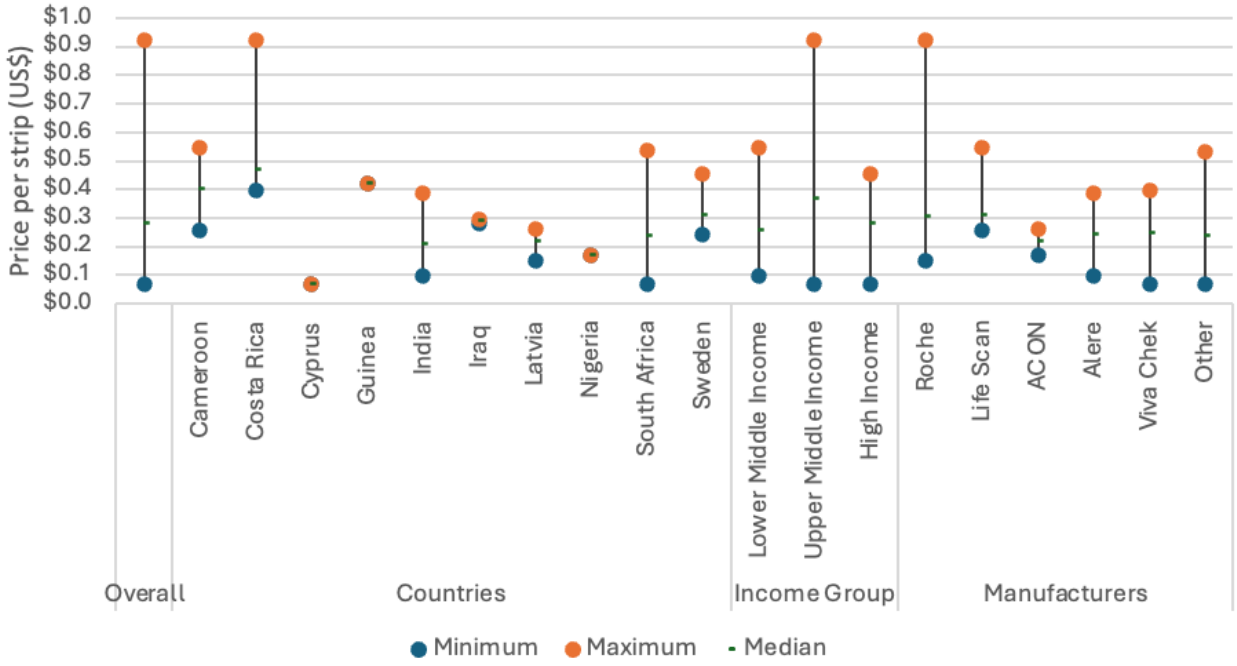
Data provided by key opinion leaders on pack size, facility type was often missing or hard to interpret. Therefore, only manufacturer, brand and price data were analysed. Countries providing data that could be used were: Cameroon; Canada; Costa Rica; Cyprus; Guinea; India; Iraq; Latvia; Nigeria; South Africa; and Sweden. Eleven different manufacturers of Blood Glucose Strips were reported as follows.

Table 12. Number of mentions of different brand of BGM strips by country

Brand	Number of mentions	Countries
Roche	15	Cameroon; Costa Rica; Guinea; India (2 mentions); Iraq; Latvia; South Africa (4 mentions) ; Sweden (4 mentions)
LifeScan	4	Cameroon; India; Sweden (2 mentions)
Viva Check	4	Costa Rica; Iraq; Latvia; South Africa
CodeFree	3	Costa Rica (2 mentions); South Africa
ACON	2	Nigeria; South Africa
Alere	2	India (2 mentions)
Contour	2	Latvia; South Africa
GlucoMen LX	2	Sweden (2 mentions)
APEX	1	Cyprus
Dr. Morepen	1	India
Jan Aushadhi Kendra glucometer strips	1	India

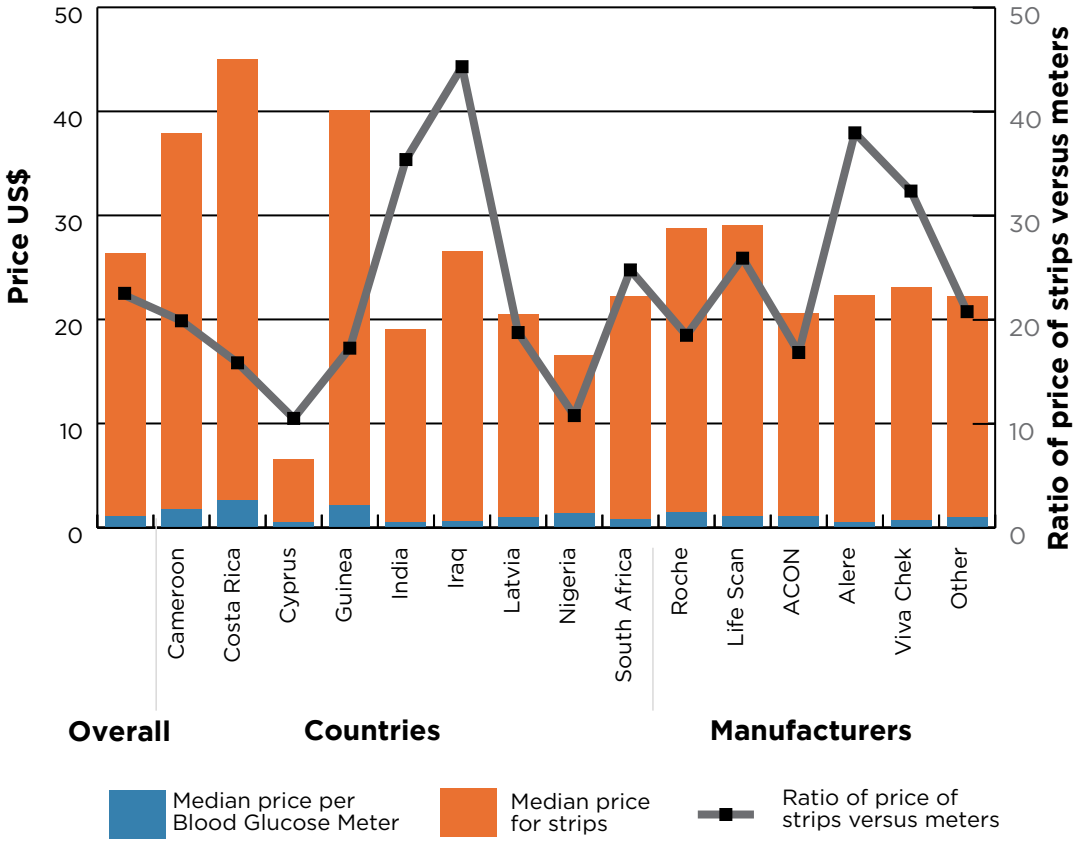
For price data 37 data points were available. As with BGMs a wide range of prices is found overall, by country and by brand (see Fig. 16). The highest priced test strip is 14 times more costly than the lowest priced strip and median prices in lower middle income countries (US$0.26) are similar to those from high income countries (US$0.28), but lower than what is reported from upper middle income countries (US$0.37).

Fig. 16. Minimum, maximum, and median prices for strips for Blood Glucose Meters as reported by key opinion leaders



Similar to BGMs the findings for strips for BGMs shows high variability as well as some manufacturers being more frequently reported on in terms of price of strips. Data from key opinion leaders found that in all cases the cost of strips far outweighed the cost of the actual meter. The overall data here clearly highlights that the cost element of the strips is higher than that of the BGM (see Fig.17).

Fig. 17. Monthly cost of Blood Glucose Meter versus Strips (assuming 3 strips per day) from key opinion leaders and ratio of price of strips to meters



3.3.2.4 Key opinion leaders – Price of lancets and syringes

Eight countries provided price data on lancets for a total of 22 prices. One price from Nigeria and the only price provided by Guinea had to be excluded as no pack size was provided. This resulted in 20 prices from 8 countries. The data reported on manufacturer was not always clear so only price data was analysed. The median price for 100 lancets was US$4.67 with a range from US$0.09 to US$42.15.

Eleven prices were reported from 7 countries. Sweden reported prices on insulin pens as well. The data reported on manufacturer was not always clear so only price data was analysed. The median price for 100 Syringes was US$12.11 with a range from US$8.89 to US$58.50.

3.3.2.5 Key opinion leaders – Price of BPMDs

Out of 7 reported prices from 6 countries by key opinion leaders, 4 of these were for Omron BPMDs. The median price for a BPMD was US$50.00 with a range from US$25.29 to US$81.51. BP machines as a one of cost are relatively high priced again show variability, overall, within countries and by brands.

3.3.2.6 Key opinion leaders – Price of different laboratory tests

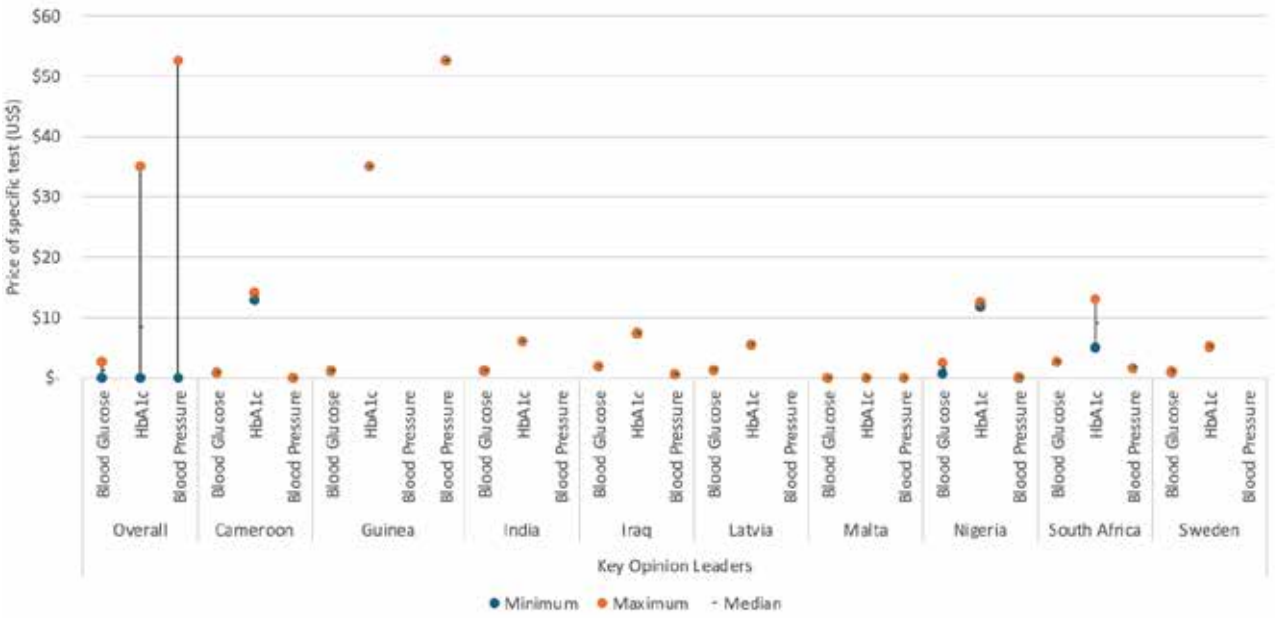
Table 13 shows the different prices that were provided by key opinion leaders with regards to the prices of different tests for patients.

Table 13. Countries and number of data points for different tests done reported by key opinion leaders

Country	Blood Glucose	HbA1c	Blood Pressure
Cameroon	2	2	2
Guinea	1	1	1
India	2	2	
Iraq	1	1	1
Latvia	1	1	
Malta	1	1	1
Nigeria	2	2	2
South Africa	1	3	1
Sweden	1	1	

In Botswana and Malawi blood glucose testing and HbA1c were reported as being free. Some respondents mentioned blood pressure measurements being free in Kenya and Nigeria. Overall, a wide range of prices for the different tests can be seen with this being pronounced in Nigeria with a wide range of prices for each test. (Figure 18) Due to lacking data no analysis could be done based on type of facility or if this was in the public or private sector. In Malta blood glucose testing and HbA1c were reported as being free. In Cameroon and Malta, and for one data point in Nigeria the measurement of blood pressure was free. Overall, a wide range of prices for the different tests can be seen. In looking at the affordability a variety of tests are free in the different countries and thus are affordable. If not, the price can be significant especially for HbA1c tests which is a cornerstone of diabetes management.

Fig. 18. Minimum, maximum, and median prices for different tests as reported by key opinion leaders



3.3.2.7 Key opinion leaders – Affordability of medical devices for the management of diabetes

In looking at the affordability based on data provided by key opinion leaders for the different medical technologies and tests as measured in number of days needed to work to afford 1 month of supplies or tests, in many countries the suggested threshold of 1 day's wage to afford 1-month of supplies is not met. (Figure 19 and Table 14)

Also, in looking at the cost of durable items, e.g., BGMs and BPMDs, these may be affordable if their lifetime cost is looked at versus people on low wages needing to disburse the overall cost of the medical technology at the time of its purchase.

Fig. 19. Affordability in days wages for the different medical technologies and tests

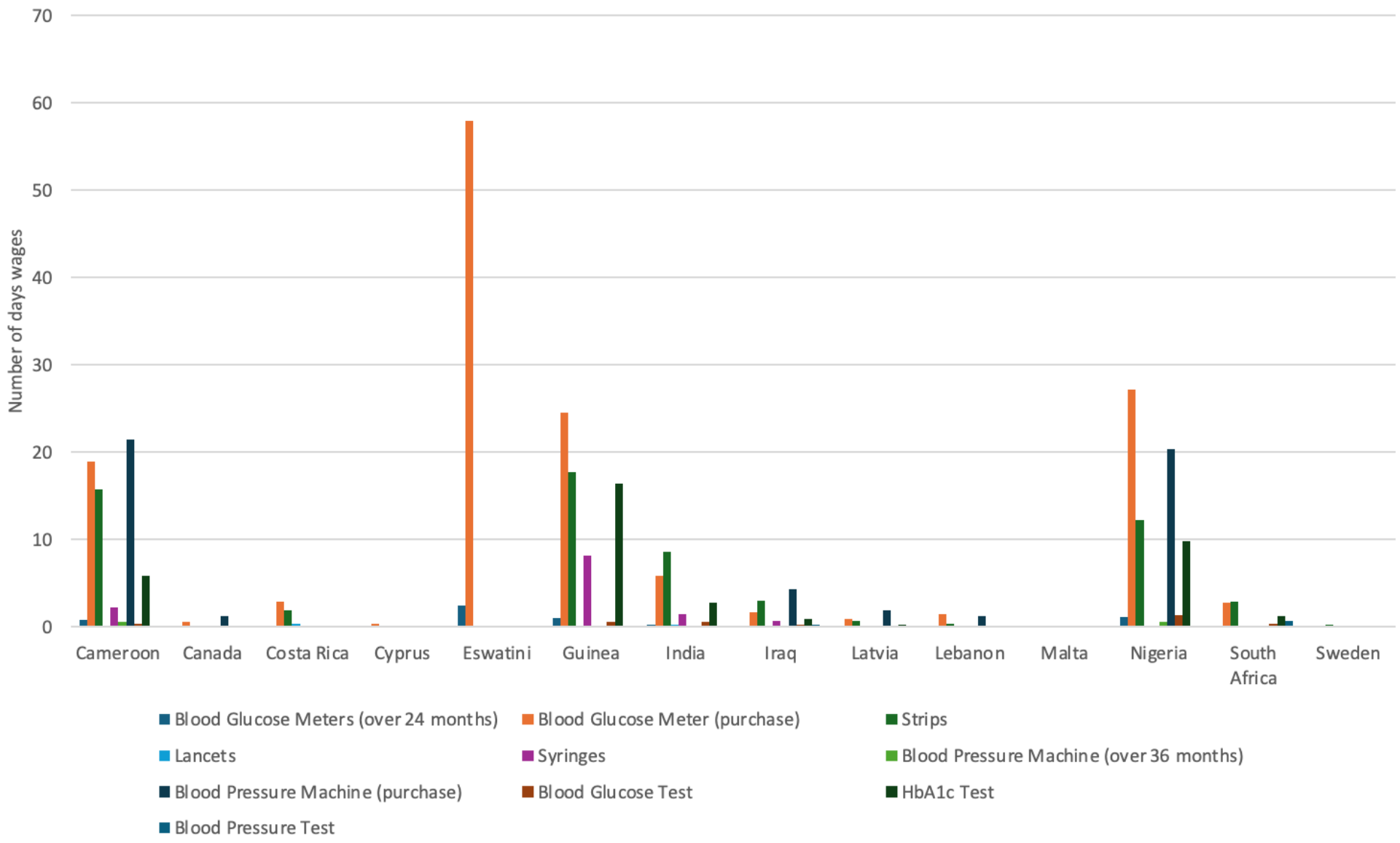


Table 14. Items per country not meeting the 1 day of wages threshold

Country	BGM (over 24 months)	Strips	Syringes	Blood Glucose Test	HbA1c Test
Cameroon		X	X		X
Costa Rica		X			
Eswatini	X				
Guinea		X	X		X
India		X	X		X
Iraq		X			
Nigeria	X	X		X	X
South Africa		X			X

In looking at the affordability of these different components of diabetes management the least affordable elements are strips and HbA1c testing meaning that many people will not be able to afford key aspects of blood glucose monitoring either on a daily basis at home or for regular follow-up in facilities.

Discussion

With regards to availability of medical devices data from the review and online survey highlight a lack of availability of the necessary medical devices for the proper management of diabetes. In looking at prices one methodological challenge is that prices of medical devices for diabetes management are reported inconsistently across the literature. From the data collected on prices the main finding is the variability of prices for blood glucose meters and strips. Although not from a representative sample, some manufacturers of BGMs and strips are more common than others. In comparing the price of a meter versus the strips the cost of the strips outweighs the cost of the actual meter. Prices for lancets and syringes were also variable although less than for BGMs and strips. The cost of for measuring HbA1c were also expensive as were the consumables. BPMDs were also found to be relatively high priced and again variability of prices, overall, within countries and by brands was found. When tests are performed in health facilities in some countries these are provided for free. If not, the price can be significant especially for HbA1c tests which is a cornerstone of diabetes management. In looking at the affordability of these different components of diabetes management the least affordable elements are strips and HbA1c testing meaning that many people will not be able to afford key aspects of blood glucose monitoring either daily at home or for regular follow-up in facilities.

Barriers identified in the literature comprise both international and national barriers. International challenges included high level barriers related to production and manufacturing, as well as accuracy and quality assurance issues. National barriers included issues within national health systems. Individual barriers include all those encountered by the individual while seeking care.

The data collected for this report sheds light on the challenges related to the availability, pricing, and affordability of medical devices necessary for diabetes management. However, it is evident that we need more comprehensive data to gain a deeper understanding of the landscape surrounding these devices. While the presented information provides valuable insights, gaps remain, calling for further research. The used methodology faced challenges, particularly due to low response rates in the survey. To enhance future data collection, we must address these limitations by improving survey design or exploring alternative methods for collecting price data. The inconsistent reporting of prices across the literature poses another methodological challenge, hindering accurate comparisons. Additional data is crucial, especially regarding differences in availability, pricing, and affordability of medical devices used both within healthcare facilities and by individuals managing diabetes at home. Furthermore, investigating overall affordability and the cost of the complete package of medicines, medical devices, and services needed for diabetes management is essential.

In trying to address the range and diversity of issues with regards to the availability, price, and affordability of different medical devices for diabetes a holistic approach is needed. A framework developed by WHO for antibiotics *(71)* and applied to insulin *(11)* can be useful to guide the next steps needed to address the challenge of medical devices (Figure 20). As the focus of this report was on availability, price and affordability some of the barriers beyond this need to be explored, for example issues surround intellectual property *(72)* as well as how medical devices are covered within the overall diabetes package within UHC to how these are delivered and supplied within the health system *(12)*.

In looking at Research & Development and Innovation two issues exist that need to be addressed. Firstly, intellectual property that might act as a barrier to improving access *(72)*. The next element is whether the innovations being developed are adapted to LMIC contexts *(73,74)*. These two factors are intimately linked to the business model of the manufacturers of medical devices. Here there needs to be a distinction between BGMs, strips, BP machines, HbA1c analysers and lancets and syringes. For BGM and strip manufacturers the element that generates the most income is the strip *(53)*. Lancets and syringes are disposable elements whereas the durable BGM and BP machine are different. The difference between a BP machine and a BGM is the issue of possibly "programmed obsolescence" with meters continually changing thus requiring new strips. In contrast BP machines can be seen as more durable in that they do not require a consumable component. HbA1c analysers might be in-between in that there might be less change in the models, but consumables/reagents are needed. A certain dominance in the global market of some manufacturers of different medical devices is present and thus competition needs to be fostered. By increasing competition regulatory bodies need to ensure safe and quality assured products. A report by the Addressing the Challenge and Constraints of Insulin Sources and Supply Study identified significant weaknesses in the regulatory pathways for BGMs *(75)*. This key step in promoting access to medical devices is needed to inform selection, pricing, and reimbursement.

Currently in most LMICs the costs of medical technologies and laboratory tests in health facilities are paid out of pocket by individuals *(12,48)*. Pricing of medical devices is relatively opaque and in many cases these need to be bought in the private sector due to little or no availability in the public sector *(27,38,42,43,55)*. This poor availability is also due to procurement and supply. Whereas insulin companies have developed differential pricing mechanisms no such initiatives exist for medical devices. It has been found that supplies of laboratory equipment and consumables in health systems is often poor *(14,27,38,39,42,43,55)*. Also for syringes it was found that there was a disconnect between the quantity of insulin ordered and syringes *(39)*. Procurement and supply need to think of the full diabetes package needed both for health facilities and individuals. The final step of prescribing, dispensing and use of medical devices is relatively under documented in LMICs. Guidelines for SMBG are needed and these then need to be in line with what is prescribed either for home use or within facilities. As the model used was developed for medicines a key stakeholder missing in this model is the laboratory technician or biomedical engineers. For facility-based testing there is the need for adapted medical technologies to the context and level of the health system for laboratory tests. For optimal use of medical technologies for SMBG and general diabetes management there is the need for education as even in donation programs where many of the medical devices and laboratory tests are provided for free poor outcomes are still being seen *(76,77)*. Also, guidelines need to consider the economic reality where lancets or syringes might be used more than they should *(78)*, as well how to best use a limited number of strips.

Fig. 20. Elements that need to be addressed for improving access to medical devices

	Current challenge(s)	Possible next steps and solutions
R&D and Innovation	• Intellectual property • Innovations adapted to LMICs	• Address issues of intellectual property • Development of universal BGM and associated strip or making strips common to various brands of glucometers. • Development of new affordable technologies adapted to LMIC contexts
Manufacturing	• Business models of companies • Dominance of some multi-nationals	• Cost of production analyses • Understanding the different markets for different medical devices • Development of access strategies by manufacturers
Marketing registration	• Weaknesses in the regulatory processes for medical devices	• Regulatory reliance process • Training of regulators
Selection, pricing, and reimbursement	• Selection of medical devices • Ongoing availability of compatible BGMs and strips • High price of medical devices • How are medical devices priced • Reimbursement of medical devices	• Inclusion of self-monitoring in UHC benefit packages (case studies) • Impact of guidelines on strips use and on cost to the individual and system • Transparency of prices • Better understanding of how prices are determined for medical devices • Government or external funding needed to make these affordable in some settings and for some populations
Procurement and supply	• Poor availability • Lack of information	• Better understanding of supply chains for medical devices • Supply chains for the health system versus for individual use • Mark-ups • Linkages and availability of right strips and meters as package or kit • Availability of full package • Supply of reagents and consumables • Linking the supply of syringes to insulin
Prescribing, dispensing and use	• Poor availability • Sub-optimal use	• Better understanding of medical device prescription, dispensation, and use • Ensure availability of adapted tools for different levels of the health system • Appropriate guidelines are developed • Need for proper diabetes care including education and empowerment • Role of laboratory technician or biomedical engineers

In looking at the issue of availability, price, and affordability of medical devices there is the need to assess these factors from the perspective of the health system and the individual. The current market for BGMs makes this unaffordable for people in LMICs and new approaches to self-monitoring adapted to LMICs are needed. This might include the development of non-propietary universal meters or strips. The data presented in this report highlights the wide range of prices for different medical devices and therefore price transparency is needed. The inclusion of BGMs in WHO prequalification is an important step, but more is needed to ensure that products being used are quality assured from the appropriate regulatory agencies. From the financing angle there is the need to see how medical devices can be included in UHC packages as at current prices these are unaffordable for many people in LMICs. This aspect needs to include a link with guidelines to ensure coherence between the clinical aspects and their financial implications for the health system and individuals. To improve affordability, mechanisms need to be put in place to improve the procurement and supply of medical devices especially ensuring linkages between the durable and consumable elements. Overall, there is a need to reconsider the element of affordability for any medical device for the management of diabetes where there is a durable and consumable element. In addition, any new technology which may eliminate the need for a consumable element will need to consider the ability to pay for people in LMICs.

The Global Diabetes Compact, WHA Resolution 74.4, WHA Resolution 76.5 all mention the issue of availability and affordability of medical devices. With the target of "100% of people with type 1 diabetes have access to affordable insulin and blood glucose self-monitoring" being far from a reality in many contexts the WHO, Member States, and other partners now need to find ways to ensure improved availability and affordability of these key elements for the management of diabetes to ensure improved outcomes for the millions of people with this condition.

References

1. Worldwide trends in diabetes prevalence and treatment from 1990 to 2022: a pooled analysis of 1108 population-representative studies with 141 million participants. Zhou, Bin et al. The Lancet, Volume 404, Issue 10467, 2077 - 20931.
2. World Health Oragnization. Diabetes. 2024. https://www.who.int/news-room/fact-sheets/detail/diabetes (accessed 25 April 2023).
3. Green A, Hede SM, Patterson CC, et al. Type 1 diabetes in 2017: global estimates of incident and prevalent cases in children and adults. *Diabetologia* 2021; 64(12): 2741-50.
4. Basu S, Yudkin JS, Kehlenbrink S, et al. Estimation of global insulin use for type 2 diabetes, 2018-30: a microsimulation analysis. *The lancet Diabetes & endocrinology* 2018.
5. World Health Oragnization. WHO list of priority medical devices for management of cardiovascular diseases and diabetes. Geneva: World Health Organization, 2021. https://iris.who.int/handle/10665/341967
6. World Health Oragnization. Priority Medical Devices Information System. 2024. https://medevis.who-healthtechnologies.org/ (accessed 4 March 2024).
7. World Health Oragnization. Electronic WHO Model List of Essential In Vitro Diagnostics. 2024. https://edl.who-healthtechnologies.org (accessed 12 January 2024).
8. Yadav H, Shah D, Sayed S, Horton S, Schroeder LF. Availability of essential diagnostics in ten low-income and middle-income countries: results from national health facility surveys. *The Lancet Global health* 2021; 9(11): e1553-e60.
9. Fleming KA, Horton S, Wilson ML, et al. The Lancet Commission on diagnostics: transforming access to diagnostics. *Lancet* 2021; 398(10315): 1997-2050.
10. Beran D, Ewen M, Laing R. Insulin in 2016: Challenge and constraints to access. *Diabetes Res Clin Pract* 2016; 117: 119-21.
11. Beran D, Lazo-Porras M, Mba CM, Mbanya JC. A global perspective on the issue of access to insulin. *Diabetologia* 2021.
12. Abraimova A, Besancon S, Portocarrero J, et al. Management of type 1 diabetes in low- and middle-income countries: comparative health system assessments in Kyrgyzstan, Mali, Peru and Tanzania. *Diabet Med* 2022: e14891.
13. Cardenas MK, Perez-Leon S, Singh SB, et al. Forty years after Alma-Ata: primary health-care preparedness for chronic diseases in Mozambique, Nepal and Peru. *Glob Health Action* 2021; 14(1): 1975920.
14. Beran D, Yudkin JS, de Courten M. Access to care for patients with insulin-requiring diabetes in developing countries: case studies of Mozambique and Zambia. *Diabetes Care* 2005; 28(9): 2136-40.
15. Beran D, Yudkin JS. Looking beyond the issue of access to insulin: what is needed for proper diabetes care in resource poor settings. *Diabetes Res Clin Pract* 2010; 88(3): 217-21.
16. World Health Oragnization. The Global Diabetes Compact. Geneva: World Health Organization, 2021. https://www.who.int/initiatives/the-who-global-diabetes-compact (accessed 4 March 2024).
17. World Health Organization. 2021. Resolution WHA A74/A/CONF./5: Reducing the burden of noncommunicable diseases through strengthening prevention and control of diabetes (24 May 2021).
18. World Health Organization 2023. Resolution WHA 76.5: Strengthening diagnostics capacity (30 May 2023).
19. World Health Oragnization. Discussion Note: Background to the WHO asks Strengthening the commitment and contribution of the pharmaceutical industry (medicines & technology) to the NCD response. Geneva: World Health Organization, 2021.
20. World Health Oragnization. Glucose Monitoring Devices - Coming Soon to PQT. 2021. https://extranet.who.int/prequal/news/glucose-monitoring-devices-coming-soon-pqt (accessed 7 February 2023).
21. World Health Oragnization. WHO releases new list of essential diagnostics; new recommendations for hepatitis E virus tests, personal use glucose meters. 2023. https://www.who.int/news/item/19-10-2023-who-releases-new-list-of-essential-diagnostics--new-recommendations-for-hepatitis-e-virus-tests--personal-use-glucose-meters (accessed 7 February 2023).
22. World Health Oragnization. First-ever global coverage targets for diabetes adopted at the 75th World Health Assembly. 2022. https://www.who.int/news-room/feature-stories/detail/first-ever-global-coverage-targets-for-diabetes-adopted-at-the-75-th-world-health-assembly (accessed 30 August 2022).
23. World Bank. World Bank Country and Lending Groups. 2024. https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups (accessed 17 January 2024).
24. Ogle GD, Middlehurst AC, Silink M. The IDF Life for a Child Program Index of diabetes care for children and youth. *Pediatr Diabetes* 2016; 17(5): 374-84.
25. Murray S, de Courten M, Raab R, Zimmet P. Is access to insulin, insulin supplies and education a democratic right for all people affected by diabetes? *Diabetes Research and Clinical Practice - DIABETES RES CLIN PRACT* 2000; 50: 337-8.
26. International Diabetes Federation Europe. Access to quality medicines and medical devices for diabetes care in Europe. Brussels: International Diabetes Federation Europe, 2013.
27. International Insulin Foundation. Report of the International Insulin Foundation on the Rapid Assessment Protocol for Insulin Access in Zambia. London: International Insulin Foundation, 2004.
28. Klatman EL, Jenkins AJ, Ahmedani MY, Ogle GD. Blood glucose meters and test strips: global market and challenges to access in low-resource settings. *The lancet Diabetes & endocrinology* 2019; 7(2): 150-60.
29. Jonsson PM, Marke LA, Nystrom L, Wall S, Ostman J. Excess costs of medical care 1 and 8 years after diagnosis of diabetes: estimates from young and middle-aged incidence cohorts in Sweden. *Diabetes Res Clin Pract* 2000; 50(1): 35-47.

30. Lee CM, Colagiuri R, Magliano DJ, et al. The cost of diabetes in adults in Australia. *Diabetes Res Clin Pract* 2013; 99(3): 385-90.

31. Hellmund R, Weitgasser R, Blissett D. Cost calculation for a flash glucose monitoring system for UK adults with type 1 diabetes mellitus receiving intensive insulin treatment. *Diabetes Res Clin Pract* 2018; 138: 193-200.

32. Tharkar S, Devarajan A, Kumpatla S, Viswanathan V. The socioeconomics of diabetes from a developing country: a population based cost of illness study. *Diabetes Res Clin Pract* 2010; 89(3): 334-40.

33. Mota-Oropeza ME, Bartley HE, Hernandez-Perez ND, et al. Providing quality care for children and adolescents with diabetes from lower-income families in Mexico. *J Pediatr Endocrinol Metab* 2019; 32(10): 1139-46.

34. Sorensen M, Arneberg F, Line TM, Berg TJ. Cost of diabetes in Norway 2011. *Diabetes Res Clin Pract* 2016; 122: 124-32.

35. Sosa-Rubi SG, Dainelli L, Silva-Zolezzi I, et al. Short-term health and economic burden of gestational diabetes mellitus in Mexico: A modeling study. *Diabetes Res Clin Pract* 2019; 153: 114-24.

36. Pfiester E, Braune K, Thieffry A, Ballhausen H, Gajewska KA, O'Donnell S. Costs and underuse of insulin and diabetes supplies: Findings from the 2020 T1International cross-sectional web-based survey. *Diabetes Res Clin Pract* 2021; 179: 108996.

37. Cavan D, da Rocha Fernandes J, Huang Y, Makaroff L. Access to Medicines and Supplies for People with Diabetes. Brussels: International Diabetes Federation, 2016.

38. Beran D, Silva Matos C, Yudkin JS. The Diabetes UK Mozambique Twinning Programme. Results of improvements in diabetes care in Mozambique: a reassessment 6 years later using the Rapid Assessment Protocol for Insulin Access. *Diabet Med* 2010; 27(8): 855-61.

39. Beran D, Atlan-Corea C, Tapia B, Martinez AJ. Report on the Rapid Assessment Protocol for Insulin Access in Nicaragua. Managua: International Insulin Foundation and Handicap International, 2007.

40. Beran D, Binh TV, Khue NT, et al. Report on the Rapid Assessment Protocol for Insulin Access in Vietnam. London: International Insulin Foundation, 2009.

41. Beran D, Silva-Matos C. Report on the Rapid Assessment Protocol for Insulin Access in Mozambique. London: International Insulin Foundation, 2009.

42. International Insulin Foundation, Santé Diabète. Final Report of the International Insulin Foundation on the Rapid Assessment Protocol for Insulin Access in Mali. London, Bamako: International Insulin Foundation and Santé Diabète, 2004.

43. International Insulin Foundation. Report of the International Insulin Foundation on the Rapid Assessment Protocol for Insulin Access in Mozambique. London: International Insulin Foundation, 2004.

44. Beran D. Key lessons learnt from the implementation of the Rapid Assessment Protocol for Insulin Access as a tool for policy change. Geneva: University of Geneva, 2014.

45. Koster I, von Ferber L, Ihle P, Schubert I, Hauner H. The cost burden of diabetes mellitus: the evidence from Germany--the CoDiM study. *Diabetologia* 2006; 49(7): 1498-504.

46. Ng CS, Lee JY, Toh MP, Ko Y. Cost-of-illness studies of diabetes mellitus: a systematic review. *Diabetes Res Clin Pract* 2014; 105(2): 151-63.

47. Lopez-Bastida J, Lopez-Siguero JP, Oliva-Moreno J, et al. Social economic costs of type 1 diabetes mellitus in pediatric patients in Spain: CHRYSTAL observational study. *Diabetes Res Clin Pract* 2017; 127: 59-69.

48. Klatman EL, McKee M, Ogle GD. Documenting and visualising progress towards Universal Health Coverage of insulin and blood glucose test strips for people with diabetes. *Diabetes Res Clin Pract* 2019; 157: 107859.

49. Ogle GD, von Oettingen JE, Middlehurst AC, Hanas R, Orchard TJ. Levels of type 1 diabetes care in children and adolescents for countries at varying resource levels. *Pediatr Diabetes* 2018.

50. Seiglie JA, Servan-Mori E, Begum T, Meigs JB, Wexler DJ, Wirtz VJ. Predictors of health facility readiness for diabetes service delivery in low- and middle-income countries: The case of Bangladesh. *Diabetes Res Clin Pract* 2020; 169: 108417.

51. Klatman EL, Ogle GD. Access to insulin delivery devices and glycated haemoglobin in lower-income countries. *World J Diabetes* 2020; 11(8): 358-69.

52. Higuchi M. Costs, availability and affordability of diabetes care in the Philippines. Tokyo: Foundation for Advanced Studies on International Development, 2009.

53. CHAI F, HAI, ACCISS,. Market report. Diabetes self-monitoring devices in low-and middle-income countries. Amsterdam: Health Action International, 2021.

54. Wambui Charity K, Kumar AMV, Hinderaker SG, Chinnakali P, Pastakia SD, Kamano J. Do diabetes mellitus patients adhere to self-monitoring of blood glucose (SMBG) and is this associated with glycemic control? Experiences from a SMBG program in western Kenya. *Diabetes Res Clin Pract* 2016; 112: 37-43.

55. Abdraimova A, Beran D. Report on the Rapid Assessment Protocol for Insulin Access in Kyrgyzstan. London: International Insulin Foundation, 2009.

56. Guy M, Gowda B, Kariithi E, et al. IDF21-0662 Diabetes CarePak: Creating & testing a bundling solution to improve access to safe insulin administration in Kenya. *Diabetes Research and Clinical Practice* 2022; 186.

57. Gabert R, Ng M, Sogarwal R, et al. Identifying gaps in the continuum of care for hypertension and diabetes in two Indian communities. *BMC Health Services Research* 2017; 17(1): 846.

58. Herkert D, Vijayakumar P, Luo J, et al. Cost-Related Insulin Underuse Among Patients With Diabetes. *JAMA internal medicine* 2019; 179(1): 112-4.

59. World Health Organization. Keeping the 100-year-old promise: making insulin access universal. Geneva: World Health Organization, 2021. https://iris.who.int/handle/10665/348384.

60. Amir S, Arash R, Kheirollah G. Low utilisation of diabetes medicines in Iran, despite their affordability (2000–2012): a time-series and benchmarking study. *BMJ Open* 2014; 4(10): e005859.

61. Wang L, Dai L, Liu H, Dai H, Li X, Ge W. Availability, affordability and price components of insulin products in different-level hospital pharmacies: Evidence from two cross-sectional surveys in Nanjing, China. *PLoS One* 2021; 16(8): e0255742.

62. Kragelund Nielsen K, Damm P, Bygbjerg IC, Kapur A. Barriers and facilitators for implementing programmes and services to address hyperglycaemia in pregnancy in low and middle income countries: A systematic review. *Diabetes Res Clin Pract* 2018; 145: 102-18.

63. Selvaraj S, Mehta A. Access to Medicines, Medical Devices and Vaccines in India. 2014: 143-59.

64. Linda M, Kanhere R, Thekkekara j. Home-use medical devices (glucometer and blood pressure monitors) -A survey of opd patients on awareness, usage, and buying behaviour determinants; 2021.

65. Utz B, De Brouwere V. "Why screen if we cannot follow-up and manage?" Challenges for gestational diabetes screening and management in low and lower-middle income countries: results of a cross-sectional survey. *BMC Pregnancy Childbirth* 2016; 16(1): 341.

66. Youngwanichsetha S, Phumdoung S. Lived experience of blood glucose self-monitoring among pregnant women with gestational diabetes mellitus: A phenomenological research. *Journal of Clinical Nursing* 2016; 26.

67. Klonoff DC, Prahalad P. Performance of Cleared Blood Glucose Monitors. *J Diabetes Sci Technol* 2015; 9(4): 895-910.

68. Manne-Goehler J, Geldsetzer P, Agoudavi K, et al. Health system performance for people with diabetes in 28 low- and middle-income countries: A cross-sectional study of nationally representative surveys. *PLoS Med* 2019; 16(3): e1002751.

69. World Health Oragnization. Assessing national capacity for the prevention and control of noncommunicable diseases: report of the 2019 global survey. Geneva: World Health Organization, 2020. https://iris.who.int/handle/10665/331452.

70. Kibriya MG, Ali L, Banik NG, Khan AK. Home monitoring of blood glucose (HMBG) in Type-2 diabetes mellitus in a developing country. *Diabetes Res Clin Pract* 1999; 46(3): 253-7.

71. World Health Oragnization. WHO report on surveillance of antibiotic consumption: 2016-2018 early implementation. Geneva: World Health Organization; 2018. https://iris.who.int/handle/10665/277359.

72. Darrow JJ, Van de Wiele V, Beran D, Kesselheim AS. An Empirical Review of Key Glucose Monitoring Devices: Product Iterations and Patent Protection. *J Diabetes Sci Technol* 2023: 19322968231178016.

73. Zafra-Tanaka JH, Beran D, Vetter B, Sampath R, Bernabe-Ortiz A. Technologies for Diabetes Self-Monitoring: A Scoping Review and Assessment Using the REASSURED Criteria. *J Diabetes Sci Technol* 2021: 1932296821997909.

74. Safary E, Beran D, Vetter B, et al. User requirements for non-invasive and minimally invasive glucose self-monitoring devices in low-income and middle-income countries: a qualitative study in Kyrgyzstan, Mali, Peru and Tanzania. *BMJ Open* 2024; 14(2): e076685.

75. Health Action International and Foundation for Innovative New Diagnostics. Regulatory Profile for Glucose Self-Monitoring Tools. Amsterdam: Health Action International, 2021.

76. Katte JC, Lemdjo G, Dehayem MY, et al. Mortality amongst children and adolescents with type 1 diabetes in sub-Saharan Africa: The case study of the Changing Diabetes in Children Programme in Cameroon. *Pediatr Diabetes* 2021.

77. Najem S, Majaliwa ES, Ramaiya K, Swai ABM, Jasem D, Ludvigsson J. Glycemic control and complications of type 1 diabetes among children in Tanzania. *J Clin Transl Endocrinol* 2021; 23: 100245.

78. Castro Adel R, Grossi SA. [Reuse of discarded syringes in residences of children and teenagers with diabetes mellitus]. *Rev Esc Enferm USP* 2007; 41(2): 187-95.

Annex 1. List of questions for online survey, questions for biomedical engineers

- Last name
- Role/ position
- Name of the health facility
- Country
- Type of health facility
- Sector
- Environment
- Location
- Are the following health technologies available at the health facility?
  - Blood glucose meters
  - Strips (or equivalent) for blood glucose meters
  - Lancets
  - Insulin syringes
  - Glycated haemoglobin analyser (HbA1c)
  - HbA1c analyser consumables (reagents and/or test strips
  - Blood pressure measurement automated or semiautomated devices
- Please add the information of the latest purchase orders with glucometers procured by the health facility
  - Brand
  - Model
  - Price
  - Pack size
  - Quantity
  - Currency
  - Acquisition year
- Please add the information of the latest purchase orders with strips (or equivalent) for blood glucose meters procured by the health facility
  - Brand
  - Model
  - Price
  - Pack size
  - Quantity
  - Currency
  - Acquisition year
- Please add the information of the latest purchase orders with lancets procured by the health facility
  - Brand
  - Model
  - Price
  - Pack size
  - Quantity
  - Currency
  - Acquisition year
- Please add the information of the latest purchase orders with insulin syringes procured by the health facility
  - Brand
  - Model
  - Price
  - Pack size
  - Quantity
  - Currency
  - Acquisition year

- Please add the information of the latest purchase orders with glycated haemoglobin analyser (HbA1c) procured by the health facility
  - Brand
  - Model
  - Price
  - Pack size
  - Quantity
  - Currency
  - Acquisition year
- Please add the information of the latest purchase orders with reagents and/or test strips for the glycated haemoglobin analyser (Hba1C) procured by the health facility.
  - Brand
  - Model
  - Price
  - Pack size
  - Quantity
  - Currency
  - Acquisition year
- Please add the information of the latest purchase orders with blood pressure measurement devices, with cuff (automated or semiautomated) procured by the health facility.
  - Brand
  - Model
  - Price
  - Pack size
  - Quantity
  - Currency
  - Acquisition year
- If the following testing are offered to the patient as a service at the health facility, please specify the price.
  - Blood glucose test
  - Glycated haemoglobin (HbA1c) test
  - Blood pressure measurement test

Annex 2. List of questions for online survey, questions for key opinion leaders

- Last name
- Role/ position
- Type of organization
- Please add the prices of blood glucose meters for patients
  - Brand
  - Model
  - Price
  - Pack size
  - Currency
  - Year
  - Facility type
  - Sector
- Please add the prices of strips (or equivalent) for blood glucose meters for patients
  - Brand
  - Model
  - Price
  - Pack size
  - Currency
  - Year
  - Facility type
  - Sector
- Please add the prices of lancets pricing for patients
  - Brand
  - Model
  - Price
  - Pack size
  - Currency
  - Year
  - Facility type
  - Sector
- Please add prices of insulin syringes for patients
  - Brand
  - Model
  - Price
  - Pack size
  - Currency
  - Year
  - Facility type
  - Sector
- Please add the prices of blood pressure measurement (automated or semiautomated) devices for patients
  - Brand
  - Model
  - Price
  - Pack size
  - Currency
  - Year
  - Facility type
  - Sector

- If you have information on prices for testing services for patients at a specific health facility, please specify the cost below
  - Blood glucose test
  - Glycated haemoglobin (HbA1c) test
  - Blood pressure measurement test

Annex 3. List of publications compiled by this scoping review

	Title	Author	Year of publication	Countries or population	Topic of interest for the scoping review	Type of Diabetes	Methodology
1	**Insulin Availability Among International Diabetes Federation Member Associations Report of the Task Force on Insulin Distribution**	Deeb	1994	60 countries	Mainly insulin. Availability of syringes 42% of the countries. Glucometers were used in 49 countries (81%).	DM	Survey
2	**Home monitoring of blood glucose (HMBG) in Type-2 Diabetes mellitus in a developing country**	Kibriya	1999	Bangladesh	Home Monitoring of Blood Glucose pack. Glucometer cost: 131.3 USD Test strips cost: 12.62 for 25 strips (1 strip 0.5 USD), 100 strips 50.5 USD (1 USD = 49.5 Taka).	T2DM	Cost-effectiveness evaluation
3	**Excess costs of medical care 1 and 8 years after diagnosis of diabetes: estimates from young and middle-aged incidence cohorts in Sweden**	Jonsson	2000	Sweden	Excess cost. Use. Annual cost strips. Strips' cost exceeds the cost of insulin. The use of blood glucose test strips was $321 in women compared to $177 USD in men	T1DM and T2DM	Cohort study
4	**Is access to Insulin, Insulin supplies and Education a Democratic Right for All People Affected by Diabetes?**	Murray	2000	16 countries Central and Eastern Europe	Insulin and glucose strip cost. 50 strips 14-57 USD. Cost per strip 0.28–1.14 USD. Cost Barriers: access in rural areas was a concern, cost, supply shortages, transportation difficulties, lack of education for patients, lack of trained personnel. Syringes and pens were reported to be available for free. Insulin cost was 5-18 USD.	DM	Survey
5	**Final report of the International Insulin Foundation on the rapid assessment protocol for insulin access in Zambia**	International Insulin Foundation	2004	Zambia	Syringe cost $0.15-1.15 USD. Glucometer availability 42-61% of facilities. Strips availability 17-32% of facilities. Cost per strip $0.58-0.74 USD. Blood glucose test cost $1.06-51.06 USD per month. Barriers: long distance to care and additional travel expenses, health workers and patients lack knowledge of diabetes. Recommendations: Improve acquisition process to ensure lowest possible cost. Relative success of DAZ Diabetes Association of Zambia increasing access to diabetes care (RAPIA)	T1DM	Report

	Title	Author	Year of publication	Countries or population	Topic of interest for the scoping review	Type of Diabetes	Methodology
6	**Final report of the international insulin foundation on the rapid assessment protocol for insulin access in Mali**	International Insulin Foundation	2004	Mali	Glucometer availability 16-100% availability. Cost of glucose test for the patient 0.08–3.00 Euro. Barriers: unavailability of diagnostics tools and the appropriate level of care, supply issues for reagents and other laboratory materials. Recommendations: removing 5% VAT and mark-ups for technologies needed for diabetes management, develop a national priority devices list for each level within the health system, develop supply chain for laboratory materials.	T1DM	Report
7	**Report of the international insulin foundation on the rapid assessment protocol for insulin access in Mozambique**	International Insulin Foundation	2004	Mozam-bique	Glucometer availability 21% of facilities. Strips availability 6% of facilities. Blood glucose test fee 0.21 USD. Barriers: unavailability of diagnostics tools at the Hospital, lack of training for healthcare providers, self-monitoring equipment out of reach for patients. Recommendations: Improve access to diagnostics tools, develop diabetes care guidelines, and improve training for laboratory technicians.	T1DM	Report
8	**Assessing health systems for type 1 diabetes in sub-Saharan Africa: Developing a 'Rapid Assessment Protocol for Insulin Access'**	Beran	2006	Mozam-bique, Mali and Zambia	System barriers. Availability and pricing of insulin, syringes and monitoring equipment. Levels of provision in the system classified as macro, meso, and micro (RAPIA)	T1DM	Assessment development and piloting
9	**Diabetes care in sub-Saharan Africa**	Beran	2006	Africa	Mainly insulin cost and availability. Glucometer availability in Mozambique 18%. Recommendations. RAPIA	DM	Review
10	**The cost burden of diabetes mellitus: The evidence from Germany - The CoDiM study**	Köster	2006	Germany	Excess cost. Annual cost. Excess cost on only medical devices 229 Euros.	T1DM and T2DM	Secondary data analysis
11	**Report on the rapid assessment protocol for insulin access in Nicaragua**	Beran	2007	Nicaragua	Barriers: availability and supply issues with test strips reagents and other devices for diagnostics. High cost of self-monitoring equipment. Strip cost 0.20 USD. Glucometer cost 120 USD. Blood glucose test cost 2.5 USD. Brands of available glucometers: ACON and Roche. Glucometer availability 95% of facilities. Strips availability 72% of the facilities. Recommendations: Improve supplies mechanisms for strips and reagents, development of a diabetes toolkit, limit number of available glucometer brands in the public sector to limit the type of strips needed, provide education to health workers regarding the use of self-monitoring equipment.	T1DM	Report

	Title	Author	Year of publication	Countries or population	Topic of interest for the scoping review	Type of Diabetes	Methodology
12	**Measuring medicine prices, availability, affordability and price components**	World Health Organization	2008	NA	Methodology for medicines that can be adapted for medical devices. Measuring price components.	NA	Survey questionnaire and detailed methodology
13	**Report on the rapid assessment protocol for insulin access in Kyrgyzstan**	Abdraimova	2009	Kyr-gyzstan	Barriers: scarcity of laboratory tools, reagents and consumables mainly due to financial constraints. Glucometer availability 67% of facilities. Strip availability 38% of facilities. Recommendations: to improve availability of diagnostic tools, define set of diagnostic tools for each level of care within the health system.	T1DM	Report
14	**Report on the rapid assessment protocol for insulin access in Vietnam**	Beran	2009	Viet Nam	Monthly cost SMBG 27 USD. Cost per strip 0.49 USD. Cost of Glucometer 87.03 USD. Fee for glucose test 1.68 USD. Also includes cost of syringe, HbA1c and lancet. Reported use of glucose testing 1 test per week to 14 tests per week. Barriers, transportation cost, lack of insurance, lack of understanding of reimbursement scheme. Recommendations: remove VAT, duties and other mark ups for medicines and devices; improve education for patients and health workers, information campaigns on health insurance schemes and relevant benefits (RAPIA).	T1DM	Report
15	**Report on the rapid assessment protocol for insulin access in Mozambique**	Beran	2009	Mozam-bique	Patient cost for blood glucose test 1.88 USD. Glucometer cost 104.86 USD, Strip cost 0.59 USD. Glucometer availability 87% of facilities. Strips availability 27%. Syringe price. Barriers: lack of devices at the health centre, greater access challenges for people living further from the health facility. Recommendations: Standardization of glucometers in the public sector, development of a toolkit for detections of diabetes complications, discussion with the public sector to guarantee availability of certain glucometers and strips.	T1DM	Report
16	**Costs, availability and affordability of diabetes care in the Philippines Foundation for Advanced Studies on International Development**	Higuchi	2009	Philippines	RAPIA questionnaires including: Interview to the health office, Interview to hospital, Interview to health centre, Interview to Pharmacy, Interview to Laboratory, Interview healthcare providers, and Interview diabetes patient. Barriers: price of the medicine and test is the most serious barrier. Self-monitoring device and strips are way beyond affordability. Strip cost 17.4 USD	DM	Country assessment

	Title	Author	Year of publication	Countries or population	Topic of interest for the scoping review	Type of Diabetes	Methodology
17	**Looking beyond the issue of access to insulin: What is needed for proper diabetes care in resource poor settings**	Beran	2010	5 countries. Mali, Mozambique, Nicaragua. Viet Nam. Zambia	Barriers: problems with distribution, tendering, policies, lack of availability of testing equipment. Blood glucose test fee Mali 2.38 USD, Mozambique 0.21USD, Nicaragua 0 USD. Percentage of income spent in diabetes care: Mali 61%, 2004; Mozambique 75%, 2003; Nicaragua 7%, 2007; Zambia 21%, 2003; Viet Nam 51%, 2008. Syringes prices. Availability of glucometers in health facilities: Mali 43%, Mozambique 21%, Nicaragua 95%, Viet Nam 100%, Zambia 54%.	T1DM and T2DM	In country assessment
18	**The Diabetes UK Mozambique Twinning Programme. Results of improvements in diabetes care in Mozambique: A reassessment 6 years later using the Rapid Assessment Protocol for Insulin Access**	Beran	2010	Mozambique	Availability of glucometers 2003 21%, 2009 87% of facilities and strips availability in 6% 2003, 27% 2009 of the facilities. RAPIA	DM	Country assessment
19	**The socioeconomics of diabetes from a developing country: A population based cost of illness study**	Tharkar	2010	India	Annual expenditure for a patient including direct 525.5USD and indirect costs 102.8 USD. Annual expenditure in India 2010 31.9 Billion USD. Annual cost for SMBG 53.7 USD	DM	Retrospective population based, cost of illness study
20	**The insulin dilemma in resource-limited countries. A way forward?**	Gill	2011	NA	Mainly on insulin. Pen vs. injection cost. Recommendation provision of diabetes education, glucose self-monitoring and expert healthcare providers as the package of care	DM	Commentary
21	**Access to quality medicines and medical devices for diabetes care in Europe**	International Diabetes Federation Europe	2013	47 countries in Europe	Definition of: availability, accessibility and affordability. Country profiles with coverage for glucometer and other devices by country. Retail pricing of strips by sub-region: West 37.3 USD, Mediterranean 42.84 USD, Central 20.50 USD South-East 25.71 USD, and Independent 27.19 USD. Strip cost to individual by sub-region: West 0 USD, Mediterranean 0 USD, Central 1.34 USD, South-East 10.71 USD, and Independent 25.84 USD.	DM	Report

	Title	Author	Year of publication	Countries or population	Topic of interest for the scoping review	Type of Diabetes	Methodology
22	**Levels of type 1 diabetes care in children and adolescents for countries at varying resource levels**	Lee	2013	Australia	Annual direct cost, $4390 AUD. Annual cost for glucometer and strips (5.2% ~ 228 AUD), In 2005 total cost for diabetes in Australia $10.6 billion AUD, annual excess $4.5 Billion AUD. Glucometer was assumed to be renewed every 3 years. The calculated cost was for the glucometer was not specified in the paper	DM	Secondary data analysis
23	**Key lessons learnt from the implementation of the Rapid Assessment Protocol for Insulin Access as a tool for policy change**	Beran	2014	Mozam-bique, Zambia, Nicaragua, Viet Nam, Kyr-gyzstan	Glucometer availability at health facilities (RAPIA). Mozambique (2003) 21%, Annual testing cost (2.73 USD) Mozambique (2009) 87% Zambia (2003) 54% Annual testing cost (11.95 USD) Mali (2004) 43%, Annual testing cost (27.15 USD) Nicaragua (2006) 95% Annual testing (0.0 USD) Viet Nam (2008) 96%, Annual testing cost (21.35 USD) Kyrgyzstan (2009) 67%	DM	Report
24	**Cost-of-illness studies of diabetes mellitus: A systematic review**	Ng	2014	30 articles	Cost of illness. Annual cost. Economic impact of DM. Excess Cost. Estimates for the total annual costs of DM ranged from US$141.6 million to US$174 billion; direct costs ranged from US$150 to US$14,060 per patient per year (pppy) whereas indirect costs ranged from US$39.6 to US$7,164 pppy	T1DM and T2DM	Review
25	**Low utilisation of diabetes medicines in Iran, despite their affordability (2000-2012): a time-series and benchmarking study**	Sarayani	2014	Iran (Islamic Republic of)	Mainly insulin. Barriers to insulin use include limited availability and affordability of syringes/needles for insulin injection or the blood glucose monitoring tests required for SMBG.	DM	Retrospective assessment
26	**Access to Medicines, Medical Devices and Vaccines in India**	Selvaraj	2014	India	Overall access to medical devices (no mention of glucometers specifically). OOP expenditure on diagnostics increasing from 2.1% in 1994 to 7.6% in 2011-2012. Barriers: use of inefficient unregulated and expensive medical devices. Access increase only for the population that can pay.	NA	Report
27	**Performance of Cleared Blood Glucose Monitors**	Klonoff	2015	NA	Barriers. A proportion of cleared BGMs do not perform at the level for which they were cleared or according to international standards of accuracy. Clinical and economical risks of inaccuracy in blood glucose measurements.	NA	Review

	Title	Author	Year of publication	Countries or population	Topic of interest for the scoping review	Type of Diabetes	Methodology
28	**Trends in Medicaid Reimbursements for Insulin From 1991 Through 2014**	Luo	2015	USA	Annual cost of reimbursement (mainly insulin).	DM	Secondary data analysis
29	**Constraints and challenges in access to insulin: A global perspective**	Beran	2016	19 countries	Availability. Affordability. Cost. Barriers mainly to insulin access	T1DM and T2DM	Review
30	**International Diabetes Federation. Access to Medicines and Supplies for People with Diabetes.**	Cavan	2016	82 Countries	Cost of glucometers and strips as percentage of income by country income level. Other devices: syringes and needles, Insulin pens. Recommendations: reducing price, improving education, improving evidence and monitoring, investing in the health system, health insurance schemes. Availability vs. availability free of charge.	DM	Access survey
31	**The IDF Life for a Child Program Index of diabetes care for children and youth**	Ogle	2016	71 countries	Government provision of glucose test strips by country income level. Annual cost of 2 strips per day $219-$730 (cost per strip 0.6–1.0 USD). Methods: Includes questionnaire. Additional info on syringes and Hb1Ac.	T1DM	Survey
32	**Cost of diabetes in Norway 2011**	Sorensen	2016	Norway	Excess cost in 2011 in Norway was 516 million Euros. BGM strips expenditure was 55 million Euros. Insulin expenditure 45.1 million Euros	DM	Secondary data analysis.
33	**"Why screen if we cannot follow-up and manage?" Challenges for gestational diabetes screening and management in low and lower-middle income countries: results of a cross-sectional survey**	Utz	2016	26 LLMIC	Barriers: financial cost (including glucometers and test strips), ability to pay for transport and treatment, fees associated with testing, shortage of laboratory equipment, calibration of the glucometer, scarce supplies.	GDM	Cross sectional survey of healthcare providers
34	**Do diabetes mellitus patients adhere to self-monitoring of blood glucose (SMBG) and is this associated with glycemic control? Experiences from a SMBG program in western Kenya**	Wambui	2016	Kenya	Adherence to SMBG. Glucometer subsidized cost $5.90 USD. 50 strips fee 37.50 USD Strip cost 0.75 USD. Payment of strips was associated with poor adherence.	DM	Retrospective cohort study

	Title	Author	Year of publication	Countries or population	Topic of interest for the scoping review	Type of Diabetes	Methodology
35	**Identifying gaps in the continuum of care for hypertension and diabetes in two Indian communities**	Gabert	2017		Glucometers and test strips were not available in many of the lower-level public facilities. Barriers: glucometers are not required in this level of care by IPHS guidelines. Shimla-Availability of glucometers and strips 50% public facilities, 100% in private facilities. Udaipur- Availability of glucometers 100% in public and private facilities strips 100% in public and 80% in private facilities. Recommendation, to include glucometers in guidelines.	DM	Household and health facility surveys
36	**Pocketbook for management of diabetes in childhood and adolescence in under-resourced countries**	International Diabetes Federation	2017	NA	Chapter 6 dedicated to blood glucose monitoring in under-resourced settings.	T1DM T2DM	Clinical Guidelines
37	**Social economic costs of type 1 diabetes mellitus in paediatric patients in Spain: CHRYSTAL observational study**	López-Bastida	2017	Spain	Annual cost for patient with T1DM was 27,274 Euros, direct cost was 4,070 Euros. Informal (familial) care represented 83% of the total cost. Average cost per patient for medical materials (including glucometers) 2,034.7 Euros.	T1DM	Cross section-al observation-al study
38	**Lived experience of blood glucose self-monitoring among pregnant women with gestational diabetes mellitus: phenomenological research**	Youngwan-ichse-tha	2017	Thailand	Barriers. Anxiety for finger prick to collect the blood by themselves, afraid of pain from the prick, afraid of elevated blood sugar results.	GDM	Phenomeno-logical study (qualitative data collec-tion)
39	**Insulin Access and Affordability Working Group: Conclusions and Recommendations**	Cefalu	2018	USA	Mainly insulin Access and Affordability Working Group: Conclusions and Recommendations. Price of insulin has consistently increased, nearly tripling between 2002 and 2013. Information on insulin pens and vials.	NA	Retrospective analysis of secondary data and interviews
40	**Cost calculation for a flash glucose monitoring system for UK adults with type 1 diabetes mellitus receiving intensive insulin treatment**	Hellmund	2018	The United Kingdom	In 2011 the annual Cost of SMBG in The United Kingdom was 158 million Pounds (21% of diabetes total cost). National Expenditure 2010-2011 was 1.9 billion Pounds (direct and indirect costs). Daily test estimate was 4-10. Barriers: inconvenience, invasiveness, pain, social stigma. Cost of lancets 0.04 Pounds and strips 0.29 Pounds (0.39 USD, 2018 rate 1.33USD). Cost estimates for 3 scenarios. Base case was 10 tests per day annual estimate 1205 Pounds.	T1DM	Case based cost calculation

	Title	Author	Year of publication	Countries or population	Topic of interest for the scoping review	Type of Diabetes	Methodology
41	**Barriers and facilitators for implementing programmes and services to address hyperglycaemia in pregnancy in low and middle income countries: A systematic review**	Kragelund	2018	LMICs	Barriers: lack of education on SMBG from the health provider, lack of glucometers and strips, concerns about blood loss, discomfort, anxiety, lack of information.	GDM	Review
42	**Cost-Related Insulin Underuse Among Patients with Diabetes**	Herkert	2019	USA	Insulin mainly, 25% of participants reported cost related insulin underuse, out of these percentage 29% reported changing insulin type due to cost (to cheaper option). Patients with underuse had poorer glycaemic control. Barrier: 60% of them experienced difficulties affording diabetes equipment.	T1DM and T2DM	Survey at Diabetes Centre
43	**The cost of life: Getting the insulin you need**	IDF	2019	USA, South Africa	Patient experience	T1DM	Interview
44	**Documenting and visualising progress towards Universal Health Coverage of insulin and blood glucose test strips for people with diabetes**	Klatman	2019	44	Six availability patterns. Availability presented as a percentage with the following scale (100% always, 75% mostly, 50% sometimes, 25% rarely, and 0% never available). OOP cost for strip $0.20-1.65 (median $0.49, mean $0.56 USD). Barriers: pharmacy stock-out, logistical problems, cumbersome regulatory procedures, customs and taxes. Barriers are greater in rural setting. The cost of strips can outweigh the cost of insulin. EDL does not include glucometer for self-management.	T1DM	Survey

	Title	Author	Year of publication	Countries or population	Topic of interest for the scoping review	Type of Diabetes	Methodology
45	**Blood glucose meters and test strips: global market and challenges to access in low-resource settings**	Klatman	2019	Low-re-source settings	Review focusing on utility, usage, cost and access to SMBG supplies in low resource settings Barriers are describe in a 3-level framework. International level, National level and patient level. Recommendations: glucometers and strips should be viewed similar to essential medicines, reduce mark-up, create TPPs and technical specifications, pricing arrangements, better procurement, and enhance diabetes education for provider and patient. Strip cost 2015, 0.29 USD. Glucometer cost 2015 12.90 (United Republic of Tanzania) and 44.10 USD (Netherlands (Kingdom of the)). In Mexico annual direct cost of T1DM $1690 SMBG accounting for 53%). daily cost of strips ranged from $0.40 Somalia to $2.40 Burkina Faso, mean cost $1.02 and median cost $1.00	DM	Review
46	**Providing quality care for children and adolescents with diabetes from lower-income families in Mexico**	Mota-Oropeza	2019	Mexico	Availability and use of SMBG. SMBG annual cost 895.3 USD. Treatment and monitoring cost USD 1,689.37. SMBG represented 53.0% of diabetes such cost. SMBG more expensive than human insulin. Only insulin is provided by the government.	DM	Retrospective analysis of secondary data
47	**Levels of type 1 diabetes care in children and adolescents for countries at varying resource levels**	Ogle	2019	Multiple	Levels of care concept with three tiers: “minimal care”, “intermediate care”, and “comprehensive (guidelines-based) care. “Each tier contains levels, which describe insulin and blood glucose monitoring regimens, requirements for haemoglobin A1c (HbA1c) testing, complications screening, diabetes education, and multidisciplinary care in varying resource settings. The cost of 2 blood glucose levels test a day could exceed the cost of insulin.	T1DM	Review
48	**When High Prices Mean Needless Death**	Rosenthal	2019	USA	Increasing price trends of insulin (Mainly insulin). Insulin pen in USA $140 USD vs. Germany and Canada $15 USD.	T1DM	Commentary
49	**Quality of medical products for diabetes management: a systematic review**	Saraswati	2019	NA	Quality barriers and unmet standards. 10% of SMBG and medicines were substandard or falsified products. Issues with the quality of the supplies. Strip degradation or contamination due to faulty transportation or sealing. The cost for SMBG can account one-third of the total cost of consumables, including insulin and needles, prompting falsification of glucose strips and reselling of second-hand unused glucose strips at reduced price.	NA	Systematic review

	Title	Author	Year of publication	Countries or population	Topic of interest for the scoping review	Type of Diabetes	Methodology
50	**Short-term health and economic burden of gestational diabetes mellitus in Mexico: A modelling study**	Sosa-Rubi	2019	Mexico	Glucometer, lancet and self-testing kit cost. Glucometer price 42 USD, Lancet 39 USD, Self-testing kit 0.3. Total cost of SMBG during pregnancy 151.8-174 USD. Excess cost of GDM 1043.9–1576.2 USD	GDM	Model
51	**Second WHO Model List of Essential In Vitro Diagnostics**	World Health Organization	2019	NA	Glucose meter inclusion in General IVDs for use in community settings and health facilities without laboratories	NA	WHO guidance
52	**Access to insulin delivery devices and glycated haemoglobin in lower-income countries**	Klatman	2020	41 countries	Other devices. Use of syringes (83.1%), pens (16.7), and pumps (0.2%). Syringe cost. Re-use of syringes reported in 94.3% of countries as well as infections on injection site. HbA1c availability in centres (83.8%), buying mean cost for the centre was 6.89 USD (2.20-18.501 USD), test mean cost for the patient was 9.34 USD (2.36 -18.5 USD). Methodology: questionnaires available. Barriers: cost, lack of syringes disposal, lack of supplies for SMBG, availability only in big centres, long distance for rural patients, need for refrigeration and cold chain for Hb1AC, inadequate training, lack of local supplier, stocks-out, lack of knowledge to interpret results, high import taxes and tariffs, custom clearance delays. Importance of Hb1Ac test. Recommendations: 1) provision of adequate syringes, 2) provision of pens and cartridges is even better, 3) recommendations to limit the risk of re-using syringes, 4) development of guidelines for low-resource settings, 5,6) up-scale Hb1Ac test, 7) improve Hb1Ac devices design for low-resource settings, 8) improve diabetes education for health professionals and patients, 9) promote pricing arrangements for low-resource settings (by manufacturers), 10) reduce and remove import taxes and tariffs by warranting the devices essential medical device status.	T1DM	Facility survey
53	**Predictors of health facility readiness for diabetes service delivery in low- and middle-income countries: The case of Bangladesh**	Seiglie	2020	Bangladesh	Health facility readiness and availability in LMICs for out-patient capabilities. Barriers: readiness was lowest in rural settings, identified districts with high social deprivation and public health facilities with unavailable equipment and medicine. Availability: low availability of glucometer and strips in certain facilities.	DM	Health facility survey (DHS questionnaire)

	Title	Author	Year of publication	Countries or population	Topic of interest for the scoping review	Type of Diabetes	Methodology
54	**Assessing national capacity for the prevention and control of noncommunicable diseases**	World Health Organization	2020	Global	Blood glucose measurement availability metric (Generally available, Generally not available, Don't know). Availability by WHO Region and by country income level. Questionnaire (Section 4 capacity for NCD early detection, treatment and care within the health system)	NCDs	Survey report
55	**WHO Electronic Model List of Essential In Vitro Diagnostics**	World Health Organization	2020	NA	Glucose meter inclusion in facility level: 1 (No laboratory)	NA	WHO guidance
56	**Market report. Diabetes self-monitoring devices in low-and middle-income countries**	CHAI	2021	71 Countries	Global Landscape. List of BGM main suppliers. Revenue for SMBG commodities. Barriers: lack of TPPs, product incompatibility (between strips and glucometers), phase out, quality standards and regulation, gross revenue margins (50-80%). High price setting, product mark-ups, tariffs and sale tax, need for a robust supply chain, lack of clinical guidelines, shortage of health workers, limited patient education, poor availability of glucose testing in health facilities. Cost and pricing in country. Market trends. Recommendations: Bilateral Donor support, TPP development, Market transparency improvement, access to price agreements (low-margin, high volume), alternative procurement channels, inclusion of glucometers in national health insurance schemes, diabetes service delivery strengthening, additional research on price sensitivity in LMICs, demand for casting and cost effectiveness. Price in LMICs glucose meter $9– 25, blood glucose strips $0.06–0.88. Price in HIC glucose meter $10–45, blood glucose strips $0.17–0.87.	DM	Market report
57	**Regulatory Profile for Glucose Self-monitoring Tools**	FIND	2021	USA, EU, United Republic of Tanzania, China	Definition of self-monitoring blood glucose system. The aim is to identify challenges and opportunities that arise from the current regulation of these technologies. Identify measures to secure equitable access to quality devices for people with diabetes. Recommendations regarding regulation and quality assurance.	NA	Regulatory profile

	Title	Author	Year of publication	Countries or population	Topic of interest for the scoping review	Type of Diabetes	Methodology
58	**Home-use medical devices) glucometer and blood pressure monitors)- A survey of OPD patients on awareness, usage and buying behaviour determinants**	Linda	2021	India	Use of glucometers at home. Barriers to glucometers use: distress from high glucose results. 73% of diabetic patients use a glucometer in average 8-9 times a month.	DM	Survey
59	**Costs and underuse of insulin and diabetes supplies: Findings from the 2020 T1International cross-sectional web-based survey**	Pfiester	2021	64 countries (Mainly USA, Ghana, Canada, Philippines and United Kingdom)	OOP expenses relative to health coverage. Rationing of blood glucose test strips and insulin. Web-based survey is a possible method to apply. Monthly OOP expense on devices (276.8 USD; 95% CI [236.2, 317.3); Monthly expense on blood glucose strips (45.1 USD; 95% CI [39.1, 51.1]).	T1DM	Web based survey
60	**Availability, affordability and price components of insulin products in different-level hospital pharmacies: Evidence from two cross-sectional surveys in Nanjing, China**	Wang	2021	China	Mainly insulin. Affordability and availability. Recommendations: increasing reimbursement rate, reducing prices and taxes, encouraging local manufacturers to develop and produce insulin products and provide appropriate subsidies to enhance their competitiveness with multinational companies.	NA	Cross-sectional survey
61	**WHO list of priority medical devices for management of cardiovascular diseases and diabetes**	World Health Organization	2021	NA	Glucometer description. Glucometer recommended for general priority clinical interventions for diabetes at clinical laboratory, clinical assessment, emergency care, hospitalization, intensive care and palliative care	DM	WHO guidance

	Title	Author	Year of publication	Countries or population	Topic of interest for the scoping review	Type of Diabetes	Methodology
62	**Keeping the 100-year-old promise: making insulin access universal**	World Health Organization	2021	NA	Mainly on insulin. One chapter on: Access to affordable devices to support appropriate use of insulin. Barriers: lack of supplies for glucose measurement, insufficient availability to provide standard care for PLWD. Substandard devices in the market. Customized strips by brand of the meter, frequent updates that render technology obsolete quicker. SMBG more costly than insulin. Recommendations: Addressing the access problem associated with devices and services 1) Include devices for dose administration and glucose monitoring, 2) Manage intellectual property, 3) Implement measures to avoid freemium for glucometers (free glucometer that force patients to buy strips only from that brand). Build capacity and invest in infrastructure 1) Integrated diabetes care, 2)Monitoring infrastructure for care and demand assessment 3)Strengthen capacity in LMICs, 4) Monitor substandard and falsified products, 5) Invest in local production . Support R&D on devices responding to LMICs needs (Like the Compendium). Standardization of glucometers to facilitate compatibility and interchangeability across brands.	DM	Report
63	**Management of type 1 diabetes in low and middle income countries: Comparative health system assessments in Kyrgyzstan, Mali, Peru and Tanzania**	Abdraimova	2022	Kyr-gyzstan, Mali, Peru, United Re-public of Tanzania	Country availability. Barriers: despite inclusion in coverage package not available or affordable. Access in rural vs. urban areas. Other barriers: geographical, socioeconomic, education level and traditional beliefs, need to educate the patient and adherence issues Annual cost of glucometer and strips. Strips -Kyrgyzstan ($233.6- $635.1), Mali ($341.6-$432.2), Peru ($222.1-$666.2), United Republic of Tanzania ($226.3-$666.2).	T1DM	Multilevel health system assessment
64	**A human rights-based approach to improve access to insulin and other aspects of diabetes care**	Brennan	2022	NA	Diabetes care with a human rights approach. Recommendations: integrate human rights approach into manufacturer's activities, create a clear public policy on access to medicines, ensure simple delivery channels for discount and donation schemes, disclose information on pricing and discounting arrangements. Incorporate human rights perspective in assessments of diabetes care, monitor access and affordability.	DM	Framework
65	**The DHS program. Service provision assessment survey**	DHS	2022	NA	Methods: Questions on availability of medical devices (for diabetes) in health facilities	DM	Questionnaire

	Title	Author	Year of publication	Countries or population	Topic of interest for the scoping review	Type of Diabetes	Methodology
66	**Diabetes CarePak: Creating & testing a bundling solution to improve access to safe insulin administration in Kenya**	Guy	2022	Kenya	Provision. Cost-benefit. Barriers. Availability (70% of PLWD do not own a glucometer). COVID-19 effects. Reuse of syringes.	T1DM and T2DM	Implementation research
67	**Closing the Type 1 Diabetes gap in South-East Asia through government partnership working with non-government organisations**	May	2022	Lao People's Democratic Republic, Malaysia, Viet Nam, Cambodia and Myanmar	Barriers: unavailability of blood glucose monitoring supplies, lack of transport, poverty, food insecurity, varying policies, lack of price transparency, complexity in supply chains, lack of market competition. Epidemiological trends.	T1DM	Health facility surveillance
68	**An integrated diabetes ecosystem to address patient barriers in low-income communities in Philippines**	Praveen	2022	Philippines	Example of an integrated diabetes care program with awareness, screening, diagnosis, affordable treatment, and medical education. Success story.	T2DM	Program evaluation
69	**Sharing "a.ma.dia" Experiences of how the Institution Have Faced COVID-19 Issues**	Ramamonjisoaa	2022	Madagascar	COVID-19 effects: increased number of diabetes complications, lack of oxygen, PPE, and medicines. The price of equipment increased 800%. 18% of diabetic patients with Covid-19 died. 55% of patients coming for Diabetes consultation missed treatment.	DM	Retrospective analysis
70	**MedevIS**	World Health Organization	2022	NA	Glucometer inclusion across all units	NA	WHO guidance

Annex 4. Respondent characteristics from biomedical engineers

Forty respondents listed the countries they were from. These were:

- Benin 1 respondent
- Botswana 1 respondent
- Ecuador 1 respondent
- Ethiopia 1 respondent
- India 1 respondent
- Kenya 8 respondents
- Lebanon 1 respondent
- Malawi 1 respondent
- Mexico 2 respondents
- Nigeria 22 respondents
- South Africa 1 respondent

Forty-one respondents provided information on the type of facility they were based at. Nineteen were at General Hospitals; 10 at specialized hospitals; 1 at a specialized laboratory; and 11 responded other. Of these "Other":

- Two respondents from teaching hospitals
- Two from tertiary hospitals
- One from a donor agency
- One from an NGO
- One from a Maternity
- One consultant
- One from a health implementing organization
- One "Nursing"
- One "R&D"

In terms of sector, 35 respondents reported being from the public sector and 7 from the private sector. Of the 41 answers to the environment where they are based 37 stated "urban" and 4 "rural".

Annex 5. Respondent characteristics from key opinion leaders

Fifty-eight provided information on the country they were from, with:

- Bangladesh 1 respondent
- Belgium 1 respondent
- Bosnia and Herzegovina 1 respondent
- Cameroon 10 respondents
- Canada 3 respondents
- China 1 respondent
- Costa Rica 1 respondent
- Cyprus 1 respondent
- Democratic Republic of Congo 1 respondent
- Eswatini 2 respondents
- Guinea 1 respondent
- India 6 respondents
- Iraq 1 respondent
- Jordan 1 respondent
- Kenya 1 respondent
- Kyrgyzstan 1 respondent
- Latvia 3 respondents
- Lebanon 2 respondents
- Malta 1 respondent
- Nigeria 6 respondents
- Pakistan 2 respondents
- Rwanda 2 respondents
- Saudi Arabia 1 respondent
- South Africa 2 respondents
- Sweden 3 respondents
- Switzerland 1 respondent
- United Republic of Tanzania 1 respondent
- United States of America 1 respondent

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